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Ice maker and refrigerator

Abstract

An ice maker includes an upper assembly having an upper tray that defines upper portions of a plurality of ice making chambers. The ice maker also includes a lower assembly that is located vertically below the upper assembly and configured rotate relative to the upper assembly. The lower assembly includes a lower tray that defines lower portions of the plurality of ice making chambers, and a lower support that is configured to receive the lower tray and restrict an outward expansion of the lower portions of the plurality of ice making chambers. The lower tray includes a lower tray body configured to hold a first volume of water, and a circumferential wall that extends upward from the lower tray body and is configured to hold a second volume of water above the first volume of water.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. application Ser. No. 16/511,871, filed on Jul. 15, 2019, which claims the benefit of the Korean Patent Application No. 10-2018-0142079, filed on Nov. 16, 2018, which are hereby incorporated by reference as if fully set forth herein.

TECHNICAL FIELD

(1) The present disclosure relates to an ice maker and a refrigerator.

BACKGROUND

(2) Generally, refrigerators are appliances that can be used to cool and store food items. A storage space inside the refrigerator may be cooled using cool air, and the food items may be stored in a refrigerated or a frozen state.

(3) In some cases, an ice maker may be provided in the refrigerator. For example, water can be supplied automatically from a water supply source to an ice tray to form ice pieces. In some cases, the formed ice pieces may be removed by heating the tray or by physically removing the ice pieces. Ice pieces formed in this manner typically have crescent or cubic shapes. In some cases, spherical ice may be made by the use of appropriately designed ice trays.

(4) During the ice making process, air bubbles can become trapped inside the ice, thus leading to a cloudy, opaque appearance. Allowing the air bubbles to escape during the ice making process, on the other hand, can help lead to the formation of clear, transparent ice pieces.

SUMMARY

(5) According to one aspect of the subject matter described in this application, an ice maker

includes: an upper assembly including an upper tray that defines upper portions of a plurality of ice making chambers and that defines a water receiving hole configured to receive water to the plurality of ice making chambers; and a lower assembly located vertically below the upper assembly and configured rotate relative to the upper assembly. The lower assembly includes: a lower tray that is made of a flexible material, that is configured to contact the water received through the water receiving hole, and that defines lower portions of the plurality of ice making chambers; and a lower support that is configured to receive the lower tray and that is configured to restrict an outward expansion of the lower portions of the plurality of ice making chambers. Each of the plurality of ice making chambers is configured to: based on rotation of the lower assembly to a first position relative to the upper assembly, receive water through the water receiving hole; and based on joining of the upper portions and the lower portions of the plurality of ice making chambers at a second position different from the first position, generate an ice piece within an ice making chamber among the plurality of ice making chambers. The lower tray includes: a lower tray body that defines the lower portions of the plurality of ice making chambers, the lower portions of the plurality of ice making chambers being configured to hold a first volume of water; and a circumferential wall that extends upward from the lower tray body, that is configured to, based on the lower assembly being at the first position, come in contact with and hold a second volume of water above the first volume of water, and that is configured to, based on the lower assembly being at the second position, vertically overlap at least a portion of the upper tray. The plurality of ice making chambers are configured to, based on joining of the upper portions and the lower portions of the plurality of ice making chambers at the second position, be filled with water from the first and second volumes of water.

(6) Implementations according to this aspect may include one or more of the following features. For example, the circumferential wall may include an vertical portion that extends vertically from an upper surface of the lower tray body and a curved portion that extends laterally toward a rotation axis of rotation of the lower assembly relative to the upper assembly. In some cases, a horizontal distance from a center of the ice making chamber to an outer end of the curved portion may be greater than a horizontal distance from the center of the ice making chamber to an outer end of the vertical portion. In some cases, the lower tray may also include an horizontal extension part that extends horizontally outward from an interface between the lower tray body and the circumferential wall. Also, a distance between an upper end of the vertical portion and the horizontal extension part may be greater than or equal to a distance between an upper end of the curved portion and the horizontal extension part.

(7) In some implementation, the upper tray may include a horizontal extension part and an upper tray body that extends downward from the horizontal extension part and that defines an inner surface of the upper portions of the plurality of ice making chambers, the inner surface having a hemispherical shape. In some cases, the upper tray body may include a vertical wall that defines a first portion of an outer surface of the upper tray body, the first portion extending in a vertical direction from the horizontal extension part and a curved wall that defines a second portion of the outer surface of the upper tray body, the second portion having a curved shape extending toward the rotation axis. The vertical portion of the lower tray body may be configured to, based on the lower assembly being at the first position, hold at least a portion of the second volume of water in a space between the vertical wall of the upper tray body and the vertical portion of the lower tray body, and the curved portion may be configured to, based on the lower assembly being at the first position and the second position, vertically overlap at least a portion of the curved wall of the upper tray body.

(8) In some implementations, the plurality of ice making chambers may include a first ice making chamber and a second ice making chamber that are arranged in a direction parallel to a rotation axis of rotation of the lower assembly relative to the upper assembly. At least a portion of the circumferential wall may extend vertically upward from an upper surface of the lower tray body.

The circumferential wall may include a first circumferential wall located at the first ice making chamber and a second circumferential wall located at the second ice making chamber. A vertical distance between the upper surface of the lower tray body and an upper end of the first circumferential wall may be less than or equal to a vertical distance between the upper surface of the lower tray body and an upper end of the second circumferential wall. In some cases, the circumferential wall may include a first portion that extends vertically from a first portion of an upper surface of the lower tray body by a first length and a second portion that extends vertically from a second portion of the upper surface of the lower tray body by a second length different from the first length. The lower support may define a lower support opening configured to receive the lower tray, and an upper end of the lower support may be configured to be coplanar with an upper end of the lower tray based on the lower tray being received in the lower support.

(9) In some implementations, the lower support may include a lower plate that extends horizontally, that is located at an upper end of the lower tray body, and that defines a lower support opening configured to receive the lower tray. The lower support may also include a wall portion that extends upward from the lower plate and that is configured to face the circumferential wall of the lower tray based on the lower tray being received in the lower support. The lower tray may further include a coupling protrusion that protrudes horizontally from the circumferential wall and that is configured to insert into a coupling slit defined at the wall portion of the lower support, and the coupling protrusion may be configured to, based on being inserted into the coupling slit, protrude outward of the wall portion of the lower support.

(10) In some implementations, an upper surface of the lower tray may define, based on the lower assembly being located at the first position relative to the upper assembly, a first angle with respect to a lower surface of the upper tray, and the upper surface of the lower tray may define, based on the lower assembly being located at the second position relative to the upper assembly, a second angle with respect to the lower surface of the upper tray, the second angle being less than the first angle. The first angle may be less than 10 degrees. In some cases, the upper tray is made of a flexible material, and the upper assembly may further include an upper case located vertically above the upper tray and an upper support located vertically below the upper tray and configured to couple the upper tray to the upper case, the upper support defining a plate opening configured to allow a bottom portion of the upper tray to pass through based on the upper tray being coupled to the upper case.

(11) In some implementations, the circumferential wall of the lower tray may be configured to, based on the lower assembly being at the first position, vertically overlap a first area of an outer surface of the upper tray, and the circumferential wall of the lower tray may be configured to, based on the lower assembly being at the second position, vertically overlap a second area of the outer surface of the upper tray, the second area being greater than the first area. In some cases, the upper tray and the lower tray may be made of a silicone material, and the ice maker may further include a direct current heater that contacts the upper tray and is configured to supply heat to the upper portions of the plurality of ice making chambers. In some cases, based on joining of the upper portions and the lower portions of the plurality of ice making chambers at the second position, a first water escape passage may be defined between the curved portion of the circumferential wall and the curved wall of the upper tray body. Also, based on joining of the upper portions and the lower portions of the plurality of ice making chambers at the second position, a second water escape passage may be defined between the vertical portion of the circumferential wall and the vertical wall of the upper tray body.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a perspective view illustrating an example refrigerator.
- (2) FIG. 2 is a front view illustrating an example state in which doors of the refrigerator of FIG. 1 are opened.
- (3) FIGS. 3A and 3B are perspective views illustrating an example ice maker.
- (4) FIG. 4 is an exploded perspective view of the ice maker in FIG. 3A.
- (5) FIGS. 5-9 are cross-sectional views taken along line B-B of FIG. 3A illustrating an example ice making process.
- (6) FIGS. 10A and 10B are cross-sectional views illustrating examples of ejector pins.
- (7) FIG. 11 is a perspective view illustrating an example lower ejector.
- (8) FIG. 12 is a top perspective view illustrating an upper case of the ice maker.
- (9) FIG. 13 is a bottom perspective view of the upper case of the ice maker.
- (10) FIG. 14 is a top perspective view illustrating an upper tray of the ice maker.
- (11) FIG. 15 is a bottom perspective view of the upper tray.
- (12) FIG. 16 is a side view of the upper tray.
- (13) FIG. 17 is a top perspective view illustrating an upper support of the ice maker.
- (14) FIG. 18 is a bottom perspective view of the upper support.
- (15) FIG. 19 is an enlarged view illustrating an example heater coupling part in the upper case of FIG. 12.
- (16) FIG. 20 is a top perspective view illustrating an example coupled state between an example heater and the upper case of FIG. 12
- (17) FIG. 21 is a view illustrating an example wiring of the heater.
- (18) FIG. 22 is a cross-sectional view illustrating an example upper assembly of the ice maker.
- (19) FIG. 23 is a perspective view illustrating an example lower assembly of the ice maker.
- (20) FIG. 24 is a top perspective view illustrating an example lower case of the ice maker.
- (21) FIG. 25 is a bottom perspective view of the lower case.
- (22) FIG. 26 is a top perspective view illustrating an example lower tray of the ice maker.
- (23) FIGS. 27 and 28 are bottom perspective views of the lower tray.
- (24) FIG. 29 is a side view of the lower tray.
- (25) FIG. 30 is a cross-sectional view taken along line A-A of FIG. 3A illustrating a pre-frozen state of an example ice piece.
- (26) FIG. 31 is a cross-sectional view taken along line A-A of FIG. 3A illustrating a frozen state of the ice piece.
- (27) FIG. 32 is a top perspective view illustrating an example lower support of the ice maker.
- (28) FIG. 33 is a bottom perspective view of the lower support.
- (29) FIG. 34 is a cross-sectional view taken along line D-D of FIG. 23 illustrating the example lower assembly in an assembled state.
- (30) FIG. 35 is a plan view of the lower support.
- (31) FIG. 36 is a perspective view illustrating an example coupling between a lower heater the lower support of FIG. 35.
- (32) FIG. 37 is a perspective view illustrating example wiring connected to the lower.
- (33) FIG. 38 is an example block diagram of the refrigerator.
- (34) FIG. 39 is a flowchart of an example process of making ice in the ice maker.
- (35) FIG. 40A is a schematic diagram illustrating example reference intervals for a spherical ice piece.
- (36) FIG. 40B is a graph illustrating sample heater outputs corresponding to the reference intervals of FIG. 40A.
- (37) FIG. 41 is a graph illustrating an example relationship between temperature detected by a temperature sensor and a corresponding output of the lower heater during the water supply and ice

making processes.

(38) FIG. 42 is a sequential view illustrating an example progression of ice across the reference intervals of FIG. 40A.

DETAILED DESCRIPTION

(39) Referring to FIGS. 1 and 2, a refrigerator 1 may include a cabinet 2 that defines a storage space for storing items, for example food items. In some cases, the cabinet 2 may define a refrigerating compartment 3 at an upper portion and a freezing compartment 4 at a lower portion. Various accommodation members such as a drawer, a shelf, a basket, and the like may be provided in the refrigerating compartment 3 and the freezing compartment 4.

(40) One or more doors may be provided to open and close the storage space of the refrigerator. For example, a refrigerating compartment door 5 may be provided for the refrigerating compartment 3, and a freezing compartment door 6 may be provided for the freezing compartment 4. As illustrated in FIG. 2, the refrigerating compartment door 5 may include a pair of left/right doors that are configured to swing open, and the freezing compartment door 6 may be part of a drawer that is inserted and withdrawn from the freezing compartment.

(41) The refrigerating and freezing compartments may be arranged in various alternative ways, as readily apparent to those of ordinary skill in the art. For example, the refrigerating and freezing compartments may be arranged side by side. In some cases, the freezing compartment may be positioned above the refrigerating compartment.

(42) As illustrated in FIG. 2, an ice maker 100 may be provided in the freezing compartment 4. The ice maker 100 is configured to make ice by using supplied water. As explained further below, the ice may have a spherical shape. Alternatively, the ice maker 100 may be provided in the freezing compartment door 6, the refrigerating compartment 3, or the freezing compartment door 5. An ice bin 102 may be provided to receive and store ice generated by the ice maker 100. The ice maker 100 and the ice bin 102 may be provided in an ice maker housing 101. The ice maker 100 and the ice bin 102 may be removed, for example, for servicing or replacement.

(43) The ice made by the ice maker 100 may be obtained by a user by, for example, opening the appropriate door to gain access to the ice bin 102. Alternatively, or additionally, a dispenser 7 for dispensing water and/or ice may be provided at an external side of the refrigerating compartment door or the freezing compartment door. A transfer unit may be used to transfer the ice stored in the ice bin 102 to the user via the dispenser 7.

(44) Referring to FIGS. 3A, 3B, and 4, an ice maker 100 according to one implementation is shown. As illustrated, the ice maker 100 includes an upper assembly 110 and a lower assembly 200. The lower assembly 200 may be rotatably coupled with respect to the upper assembly 110, with the upper and lower assemblies 110, 200 being designed to come together to form an ice making chamber 111 for spherical ice. The ice making chamber 111 may be formed, for example, by a lower tray that defines the shape of a lower half of the ice and an upper tray that defines the shape of an upper half of the ice. As shown, a plurality of ice making chambers 111 may be provided. For example, three or more chambers may be linearly arranged along a row. In some cases, the chambers may be provided in multiple rows that are arranged parallel to each other. Other shapes of ice, for example cubic or cylindrical among others, may be formed using a similar configuration of upper and lower assemblies but with differently shaped ice making chambers.

(45) In more detail, referring to FIGS. 3A and 3B, the ice maker 100 includes an upper assembly 110 and a lower assembly 200. As explained further below, the lower assembly 200 is configured to rotate relative to the upper assembly 110 during the ice making process.

(46) The upper assembly 110 includes an upper case 120 that defines an outer appearance and an upper tray 150 that is mounted within the upper case 120. The upper tray 150, which can be made from a flexible material such as silicone, defines the upper portion of the plurality of ice making chambers 111. For example, in the case of spherical chambers 111 designed to form spherical ice pieces, the upper hemisphere of the chambers may be defined by the upper tray 150 (with the lower

hemisphere being defined by a corresponding lower tray, as further detailed below).

(47) The upper tray **150** defines, at its upper surface, a plurality of upper tray openings **154**. An upper ejector **300** includes a plurality of corresponding protrusions that are designed to pass through the upper tray openings **154** during an ice ejection stage to thereby push downward and remove any ice pieces that may be located within the upper portions of the ice making chambers **111**. One of the plurality of upper tray openings **154** may further be configured as a water receiving hole **112**. In some cases, the water receiving hole **112** may be separately provided to the upper tray **150** in addition to the upper tray openings **154**. In either case, the water receiving hole **112** is configured to receive water from a water supply part **190**.

(48) The water supply part **190** may be a trough-like structure that is coupled to the upper assembly **110** and that is configured to receive water from a water supply source of the refrigerator. The water supply part **190** may further include a spout-like structure through which the received water flows into the ice making chambers **111**. As illustrated, the water supply part **190** can supply water through only a single opening in the upper tray **150**. However, because the plurality of ice making chambers **111**, as explained in greater detail below, are fluidically connected to one another during the water filling stage, the water received through the single opening can be distributed to all the chambers. As a result, all of the ice making chambers **111** may be filled simultaneously with water using a single water supply part **190**. In some implementations, multiple water supply parts, or alternatively a water supply part having multiple spouts, may be used to deliver water directly to more than one chamber at a time.

(49) Referring further to FIG. **4**, which shows an exploded view of the ice maker **100**, the lower assembly **200** may include a lower tray **250**, a lower support **270**, and a lower case **210**. The lower tray **250**, which can also be made from a flexible material such as silicone, defines the lower portion of the plurality of ice making chambers **111**. For example, in the case of spherical chambers **111** designed to form spherical ice pieces, the lower hemisphere of the chambers may be defined by the lower tray **250**, with the upper hemisphere being defined by the upper tray **150** as explained above.

(50) In some cases, the lower tray **250** may be formed from a silicone material that is more elastically deformable than the silicone material used to form the upper tray **150**. Therefore, by way of example, the lower tray **250** may be more easily flexed during the ice removal process compared to the upper tray **150**.

(51) A driving unit **180** may be provided to the ice maker **100**. The driving unit **180** is configured to rotate the lower assembly **200** relative to the upper assembly **110** during the ice making process. The driving unit **180** may include a driving motor and a power transmission part, such as one or more gears, to actuate the lower assembly **200**. The driving motor may be rotatable in both directions, thereby allowing the lower assembly **200** to be rotated in both directions. Although FIG. **4** shows a single driving unit **180** provided at one side of the ice maker **100**, multiple driving units may be provided. For example, driving units may be provided at opposing sides of the ice maker.

(52) FIG. **4** further shows the upper ejector **300**, which may be removably coupled to the upper assembly **110**. The upper ejector **300** may include an ejector body **310** and a plurality of upper ejecting pins **320** that extend downward from the ejector body **310** toward the ice chambers **111**. The number of upper ejecting pins **320** provided on the ejector body **310** may correspond to the number of ice chambers **111** such that each ejecting pin is configured to be pushed downward into a corresponding ice chamber during the ice ejection stage. One or both side ends of the upper ejector **300** may include a retaining member **312** that is configured to prevent a connection unit **350** from becoming uncoupled from the upper ejector **300**.

(53) The connection unit **350**, which may include one or more links that couple the lower assembly **200** to the upper ejector **300**, is configured to translate a rotational movement of the lower assembly **200** to an up-down movement of the upper ejector **300**.

(54) For example, when the lower assembly **200** rotates in one direction, the upper ejector **300** may

descend by the connection unit **350** to allow the upper ejector pin **320** to move downward and push out the ice. Conversely, when the lower assembly **200** rotates in the opposite direction, the upper ejector **300** may ascend back to its original position.

(55) The ice maker **100** may also include a lower ejector **400** that is configured to remove ice that may be retained within the lower portion of the ice chamber **111** in the lower assembly **200**. The lower ejector **400** may include an ejector body **410** and a plurality of lower ejecting pins **420** that generally extend in a lateral and downward direction. The lower ejector **400** may be attached to the upper case **120** at a location such that, in use, when the lower assembly **200** is rotated away from the upper assembly **110**, the lower assembly **200** is actuated toward the lower ejector **400** such that the lower ejecting pins **420** can press and deform the lower tray **250** to thereby remove ice that is retained in the lower portion of the chamber **111**.

(56) As illustrated in FIG. 4, the upper assembly **110** includes the upper case **120** that holds the upper tray **150** and further includes an upper support **170** that is configured to secure the upper tray **150** to the upper case **120**. Portions of the upper tray **150**, for example, may be positioned between the upper case **120** above and the upper support **170** below to provide a more secure coupling. Various coupling features, such as bosses, fasteners, hooks, tabs, bolts, protrusions, and the like, may be provided to help couple the upper case **120**, the upper tray **150**, and the upper support **170** to each other in a vertically aligned configuration. The water supply part **190** may be attached to the upper case **120**.

(57) The ice maker **100** may also include a temperature sensor **500** for detecting a temperature of the upper tray **150**. For example, the temperature sensor **500** may be mounted on the upper case **120** such that, when the upper tray **150** is fixed to the upper case **120**, the temperature sensor **500** contacts the upper tray **150**. In other cases, the temperature sensor **500** may be mounted directly to the upper tray **150**. In some implementations, one or more other temperature sensors may be provided, for example at the lower tray **250**.

(58) The lower assembly **200** may include a lower support **270** that is configured to provide support to a lower side of the lower tray **250** and a lower case **210** that is configured to provide support to an upper side of the lower tray **250**. The lower case **210**, the lower tray **250**, and the lower support **270** may be coupled to each other through one or more coupling members, including but not limited to bosses, fasteners, hooks, tabs, bolts, protrusions, and the like.

(59) The ice maker **100** may include a switch for turning the ice maker **100** on and off. For example, the ice maker **100** may be activated to make ice when a user turns on the switch **600**. That is, when the switch **600** is turned on, water may be supplied to the ice making chambers **111** of the ice maker **100**. Subsequently, the water supplied to the ice making chambers **111** can be frozen to form ice pieces that are in turn ejected from the ice making chambers **111**.

(60) An exemplary ice making process of the ice maker **100** will be detailed below with reference to FIGS. 5 to 9.

(61) Referring to FIG. 5, water **W** may be supplied to the ice making chamber **111**, which is made up of an upper chamber **152** and a lower chamber **252**, when the lower tray **250** is in a water supply position. As explained above, the water may be received through the water receiving hole **112** from the water supply part **190**.

(62) In the water supply position, which is illustrated in FIG. 5, the lower tray **250** may be rotated about a rotation axis **cl** such that the ice making chamber **111** is not completely closed. That is, the ice making chamber **111** may remain slightly open such that a preset angle is formed between a lower surface **151e** of the upper tray **150** and an upper surface **251e** of the lower tray **250**. The preset angle may be between 0 and 90 degrees. In some cases, the preset angle may be approximately 8 degrees. By leaving the ice making chamber slightly open by the preset angle while receiving the water, adjacent chambers within the ice making chamber **111** can be fluidically connected to each other. Accordingly, even if water is supplied via the water receiving hole **112** to just one of a plurality of chambers, the supplied water can be distributed to all the chambers. That

is, all the chambers can be filled by supplying water to just one of the chambers and allowing the water to overflow into the adjacent chambers.

(63) With the lower tray **250** in the water supply position, a predetermined volume of water can be supplied to the ice making chambers **111**. The predetermined volume of water may be greater than the amount of water required to create the desired ice piece. In such cases, excess water may be channeled away from the ice making chambers through one or more water escape passages that are provided by the ice making trays, as will be described further below.

(64) When the predetermined volume of water is supplied with the lower tray **250** in the water supply position, water **W** may completely fill the lower chamber **252**. Water **W** may further fill, either partially or completely, a space that is formed between the upper and lower chambers **152**, **252**. In some cases, some of the supplied water may fill a lower portion of the upper chamber **152**. Although the upper chamber **152** may not be filled with water, water that is held in the space between the upper and lower chamber **152**, **252** can subsequently be pushed into the upper chamber **152** to thereby create a fully-formed ice piece. In order to ensure that a sufficient volume of water is retained within the upper chamber **152**, the volume of water that is held between the upper and lower chambers **152**, **252** during the water supply position may be equal to or greater than the volume of water that can be held within the upper chamber **152**.

(65) As described in further detail below with respect to FIGS. **26** to **29**, the lower tray **250** may include a circumferential wall, or a retaining wall **260**, that extends vertically upward from the upper surface **251e** and that serves to contain the water that is held above the upper surface **251e**. That is, the retaining wall **260** is designed to prevent the water that is held between the upper and lower chambers **152**, **252** during the water supply step from spilling out.

(66) Referring to FIG. **6**, the lower tray **250** is shown rotated from the water supply position shown in FIG. **5** to an ice making position. For example, the driving unit **180** may rotate the lower assembly **200** toward the upper assembly **110** such that upper surface **251e** of the lower tray **250** become coplanar with the lower surface **151e** of the upper tray **150**. Through this motion, as can be seen in FIGS. **5** to **6**, the water **W** that is held between the upper and lower chambers **152**, **252** may be pushed upward into the upper chamber **152**.

(67) In some implementations, after a complete ice making chamber has been formed in this manner, the driving unit **180** may over-rotate the lower tray **250** toward the upper tray **150** by a small amount to ensure that no gaps are present between the upper and lower surfaces **251e** and **151e**. The presence of gaps in this region between the trays **250** and **150**, for instance, may result in an undesirable seam or protrusion that is formed around formed ice.

(68) When the water **W** contained within the ice making chamber freezes, ice **I** is formed as illustrated in FIG. **7**.

(69) Referring also to FIG. **6**, a lower portion of the lower tray **250** may include a deformable portion **251b** that is configured to change shape based on an outward expansion of the ice piece within the ice making chamber during ice generation. Accordingly, the volume of the ice making chamber before ice generation (i.e. before the deformable portion **251b** changes shape) may be less than the volume of the ice making chamber after ice generation (i.e. after the deformable portion **251b** changes shape). Notably, because the deformable portion **251b** is configured to more readily change its shape compared to other portions of the ice making chamber, distortion of the chamber shape caused by ice expansion may be localized to the deformable portion **251b**.

(70) In some implementations, the deformable portion **251b** may initially have a convex shape that protrudes toward a center of the ice making chamber as shown in FIG. **6**. As illustrated in FIG. **6**, filling of the chamber with water may not generate enough pressure to substantially change the convex shape of the deformable portion **251b**. However, once the water **W** within the chamber freezes, as seen in FIG. **7**, the outward expansion of the ice **I** can push out the deformable portion **251b** to take on a concave shape that protrudes away from the center of the ice making chamber. Accordingly, the transformation of the deformable portion **251b** from a first shape (e.g. convex) to

a second shape (e.g. concave) can help the ice making chamber to provide on a more spherical shape during the ice making stage. That is, the outward expansion of the deformable portion **251b** can help compensate for the outward expansion of the ice to thereby provide a final ice shape that is more spherical than would have been otherwise. The deformable portion **251b** can revert back to its original shape (i.e. first shape) after the ice piece is removed from the chamber.

(71) The lower support **270** (FIG. **4**), which may be more rigid than the lower tray **250**, includes a recess that is configured to surround and physically support the spherical portion of the lower tray **250**. Accordingly, outward expansion of the lower tray **250** during ice formation, or other unwanted shape distortions, may be restricted. In some cases, as explained below with respect to FIG. **33**, the lower support **270** may include lower openings **274** to accommodate the deformable portion **251b** of the lower tray **250**. Accordingly, the lower support **270** can allow the deformable portion **251b** to expand outward during ice formation while at the same time providing a supporting force to the remaining portions of the lower tray **250**. In some cases, the deformable portion **251b** of the lower tray **250** may be configured to be more flexible than the other portions of the lower tray, for instance by being made thinner, to facilitate transitioning between the first and second shapes.

(72) An exemplary process of ejecting the ice piece from the ice making chamber is illustrated in FIGS. **8** and **9**. In particular, after the ice piece is formed inside the chamber, the driving unit **180** may rotate the lower assembly **200** away from the upper assembly **110** to separate and open up the upper and lower ice making chambers, thereby exposing the ice piece within.

(73) During this ejection process, as illustrated in FIG. **8**, the upper ejector **300** may move downward in conjunction with the outward rotation of the lower assembly **200** such that the upper ejecting pins **320** pass through the upper tray **150** and into the ice chamber **111**, thereby pushing away any ice remaining inside the upper chamber **152**. In this way, the ice pressed by the upper ejecting pin **320** may be separated from the upper assembly **110** and collected, for example, in the ice bin **102**. In some cases, the ice piece **I** may remain adhered to the lower chamber **252**.

(74) As the lower assembly **200** continues to rotate outward away from the upper assembly **110**, as seen in FIG. **9**, any remaining ice piece **I** may fall out toward the ice bin **102** due to gravity. In some cases, the ice piece **I** may not fall out on its own and instead remain adhered to the lower ice tray **250**. The continued rotation of the lower assembly **200** away from the upper assembly **110** in such cases will cause the lower ejecting pins **420** of the lower ejector **400** to pass through the lower openings **274** of the lower support **270** to press and deform the lower tray **250**, for instance at the deformable portion **251b**, to thereby remove any ice that is retained in the lower portion of the chamber. In some cases, as shown in FIG. **9**, a distal end of the lower ejector **400** may extend past the upper surface **251e** of the lower tray **250** in order to push any remaining ice piece. In some cases, a length of the ejector pins **420** may be equal to or greater than a radius of the ice making chamber.

(75) In order to ensure that the ice piece within the chamber is properly ejected, as illustrated in FIG. **9**, the lower assembly **200** may be rotated past 90 degrees from the ice making position. In some cases, the lower assembly **200** may be rotated between 120-140 degrees from the ice making position to reach the final ice ejection position.

(76) Various exemplary implementations of the ejector pin **420** are illustrated in FIGS. **10A** and **10B**. As shown in FIG. **10A**, the ejector pin **420a** may be substantially linear in shape. The orientation angle of the ejector pin **420a** may be chosen to be generally orthogonal to the lower assembly **200** at the final ice ejection position. For example, if the lower assembly **200** is designed to be rotated 110 degrees, the ejector pin **420a** may be angled downward by 20 degrees. If the lower assembly **200** is designed to be rotated 130 degrees, the ejector pin **420a** may be angled downward by 40 degrees. Alternatively, the orientation angle of the ejector pin **420a** may be chosen to be generally orthogonal to the lower assembly **200** when a distal end **430** of the lower ejector **400** first makes contact with the lower ice tray **250**. For example, if the lower ejector **400** first makes contact with the lower ice tray **250** when the lower assembly **200** has been rotated 90

degrees from the ice making position, the ejector pin **420a** may be oriented to be substantially horizontal.

(77) In some implementations, as shown in FIG. **10B**, the ejector pin **420b** may be curved toward the rotation shaft of the lower assembly **200**. For instance, the curvature of the ejector pin **420b** may correspond to a trajectory of the lower opening **274** such that the entire length of the ejector pin **420b** may pass through the lower opening **274** without making contact with the lower support **270**. In some cases, a radius of curvature of the ejector pin **420b** may correspond to a radial distance between the rotation axis **cl** of the lower assembly **200** and the lower opening **274**.

(78) In some implementations, as illustrated in FIG. **11**, the lower ejector **400** may include ejector pins having unequal lengths. For instance, as shown, ejector pin **420d** may be longer than ejector pin **420c**, and ejector pin **420e** may be longer than ejector pin **420d**. Accordingly, during downward rotation of the lower assembly **200** in the course of ice ejection, ejector pin **420e** may contact/push the ice in the lower tray **250** first, followed by ejector pin **420d** and then ejector pin **420c**. In this way, because contact of multiple ejector pins may be staggered, peak torque required from the driving unit **180** may subsequently be reduced. This is because motor torque required to eject three ice pieces simultaneously, for instance, is less than motor torque required to eject just one piece at a time.

(79) In some cases, a length of the ejector pin may increase along a length direction of the ejector body **410**, as exemplified in FIG. **11**. That is, a length of the ejector pin at a first end of the ejector body **410** (e.g. pin **420c**) may be the shortest among all the ejector pins, and a length of the ejector pin at a second end of the ejector body **410** that is opposite the first end may be the longest (e.g. pin **420e**). In some cases, the driving unit **180** may be provided at a side of the ice maker **100** that corresponds to the first end of the ejector body **410**. That is, the first end of the ejector body associated with the shortest ejector pin may be positioned closer to the driving unit **180** than the second end of the ejector body associated with the longest ejector pin.

(80) In some cases, torque provided by the driving unit **180** may cause the lower assembly **200** to twist as it is being rotated, particularly when a portion of the lower assembly **200** encounters additional resistance from the ejector pins. In such cases, the side of the lower assembly **200** that is farther away from the driving unit **180** may rotate at a slower rate than the side that is closer to the driving unit **180**. For example, when the side of the lower assembly that is closer to the driving unit **180** has been rotated 110 degrees, for example, the opposite side farther away from the driving unit **180** may only be rotated by 100 degrees due to the twisting (i.e. wringing effect) of the lower assembly **200**. By correspondingly increasing the lengths of the ejector pins based on their distance from the driving unit **180**, for example as shown in FIG. **11**, the extra pin length may compensate for the reduced rotation in that region stemming from the twisting effect. Accordingly, a sufficient length of the ejector pin may nevertheless be inserted through the lower opening **274**, despite the twisting, in order to eject the ice.

(81) As will be understood by a skilled artisan from the disclosure herein, different shapes, sizes, and orientations of the ejector pins may be used.

(82) Referring now to FIGS. **12** and **13**, top and bottom perspective views, respectively, of the upper case **120** of the ice maker **100** according to one implementation is shown. The upper case **120** may at least partially define an outer surface of the ice maker **100** and may be mounted within the freezing compartment **4** to thereby couple the ice maker **100** to the refrigerator **1**. In some cases, the upper case **120** may be attached to the housing **101** of the freezing compartment **4**.

(83) The upper case **120** may include an upper plate **121** to which the upper assembly **110** is coupled. For example, the upper tray **150** may come in contact with and become attached to a bottom surface of the upper plate **121**. The upper tray may include an opening **123** through which a portion of the upper tray **150** can pass through. Accordingly, when the upper tray **150** is attached to the bottom surface of the upper plate **121**, a portion of the upper tray **150** may protrude upward through the opening **123**. A more secure coupling between the upper plate **121** and the upper tray

150 may be achieved as a result.

(84) Alternatively, the upper tray **150** may be positioned above the upper plate **121** such that the upper tray **150** protrudes downward through the opening **123**. The upper plate **121** may include a recess part **122** that is recessed downward from an upper surface of the upper plate **121**. The opening **123** may be defined at a bottom surface **122a** of the recess part **122**. The upper tray **150** that protrudes downward through the opening **123** may be accommodated in the recess part **122**.

(85) As seen in FIG. **13**, a heater coupling part **124**, for example a groove configured to accommodate a heater therein, may be provided to the upper plate **121**. As further explained below with respect to FIG. **20**, the heater coupling part **124** holds an upper heater that is configured to heat the upper tray **150**. In some cases, the heater coupling part **124** may be provided vertically below the recess part **122**.

(86) The upper case **120** may include installation ribs **158** and **159**, which may protrude downward from the bottom surface of the upper plate **121**. Additional pairs of ribs may be provided to the upper case **120**. The installation ribs **158** and **159** can be used to mount the temperature sensor **500** (FIG. **4**) to the upper case **120**.

(87) For example, as seen in FIG. **13**, the pair of ribs **158** and **159** may be spaced apart from each other along a direction B. Accordingly, the temperature sensor **500** may be held between the pair of installation ribs **158** and **159**.

(88) Slots **131** and **132** may be defined in the upper plate **121**. The slots may be configured to receive and be coupled to corresponding protrusions that are provided to the upper tray **150**. In some cases, the slot-protrusion relationship may be reversed (i.e. protrusions are provided to the upper plate **121** and slots are defined in the upper tray **150**). Other types of coupling structures between the upper plate **121** and the upper tray **150** may also be used.

(89) First slots **131** may be spaced apart from the second slots **131** along the direction B such that the slots are positioned on opposite sides of the opening **123**. Each of the first slots **131** may be spaced from each other along a direction A, and each of the second slots **132** may be spaced apart from each other along the direction A. The plurality of ice chambers **111** may be arranged along the direction A. Direction A may be orthogonal to direction B and further parallel to the rotation axis **cl** of the lower assembly **200**.

(90) In some cases, the first and second slots **131** and **132** may have a curved shape, for example convex with respect to the opening **123**, thus allowing a length of each of the slots to be extended. By increasing the slot length, along with the length of the corresponding protrusion of the upper tray **150**, a coupling force between the upper tray **150** and the upper case **120** may be increased.

(91) In some implementations, a distance between the first upper slot **131** and the opening **123** may be different from that between the second upper slot **132** and the opening **123**. For example, the distance between the first upper slot **131** and the opening **123** may be greater than that between the second upper slot **132** and the opening **123**.

(92) Referring to FIG. **12**, the upper plate **121** may include a plurality of sleeves **133** that are configured to receive corresponding coupling bosses **175** of the upper support **170** (FIG. **17**). The sleeve **133** may have a cylindrical shape and extend upward from the upper plate **121**. A plurality of sleeves **133** may be provided on the upper plate **121**. The plurality of sleeves **133** may be arranged to be spaced apart from each other in the direction of the arrow A. In some cases, the plurality of sleeves **133** may be arranged in a plurality of rows in the direction of the arrow B. In some cases, each of the sleeves **133** may be positioned between adjacent ones of the slots **131** and/or between adjacent ones of the slots **132**.

(93) Referring to FIG. **13**, hinge supports **135** and **136** may be provided to the upper case **120**. The hinge supports **135** and **136** may protrude downward from the bottom surface of the upper plate **121** and are configured to rotatably support the lower assembly **200**. A hinge opening **137** may be defined in each of the hinge supports **135** and **136**.

(94) Referring back to FIG. **12**, the upper case **120** may include a vertical extension part **140** that

extends vertically upward from an upper surface of the upper case **120** and further extends circumferentially around the upper plate **121**. The vertical extension part **140** may extend upward from the upper plate **121**. The vertical extension part **140** may include one or more coupling hooks **140a** that are configured to couple the upper case **120** to the housing **101**. The water supply part **190** (FIG. 4) may be coupled to the vertical extension part **140**, for example via coupling slots defined the vertical extension part **140**.

(95) The upper case **120** may further include a horizontal extension part **142** that extends horizontally outward from the vertical extension part **140** to form an upper horizontal surface of the upper case **120**. The horizontal extension part **142** may include a screw coupling part **142a** that is configured to receive a screw that couples the upper case **120** to the freezer compartment.

(96) The upper case **120** may further include a circumferential sidewall **143** that extends downward from the horizontal extension part **142** and at least partially surrounds a circumference of the upper and lower assemblies **110**, **200**. The circumferential sidewall **143** may form an eternal appearance of the ice maker **100** and helps provide a protective barrier between the various moving components of the ice maker **100**, such as the lower assembly **200**, and the rest of the freezing compartment. As illustrated in FIG. 13, one side of the circumferential sidewall **143** may be left open to, for example, allow a user to access the inside of the ice maker **100**. In some cases, the lower ejector **400** may be attached to an inner side of the circumferential sidewall **143**.

(97) Referring now to FIGS. 14 to 16, the upper tray **150** includes, among other things, the upper chamber **152** that provides a mold for shaping the upper half of the ice piece being made. The upper chamber **152** may be hemispherical in shape, for example, to form the upper hemisphere of a spherical ice piece. The upper chamber **152** may include an array of upper chambers, such as upper chambers **152a**, **152b**, **152c**, to enable making multiple ice pieces at a time.

(98) The upper tray **150** may be integrally molded as one piece. Alternatively, the upper tray **150** may be made from separate pieces that are attached together.

(99) In one implementation, the upper tray **150** may be made of a flexible material that is capable of being restored to its original shape after being deformed by an external force. For example, the upper tray **150** may be made of a silicone material. Accordingly, the upper tray **150** may be deformed during, for example, the ice ejection process but may subsequently return to its original shape to generate additional ice pieces. The spherical shape of the ice, therefore, may be maintained through repetitive uses. In some cases, the upper tray **150** may be intentionally deformed during the ice ejection process to facilitate removal of the ice piece.

(100) In some cases, for reasons discussed below, the upper tray **150** may be made from a heat-resistant material that will maintain its shape when heated. A silicone material, which exhibits good heat resistance, may also be used for this purpose.

(101) The upper tray **150** may include an upper tray body **151** that defines an internal space for molding ice, namely one or more upper chambers **152** that make up the upper half of the ice chamber **111**.

(102) In one implementation, the upper chambers **152** may include a first upper chamber **152a**, a second upper chamber **152b**, and a third upper chamber **152c**. The one or more upper chambers **152** may be defined within a chamber wall **153** that forms an outer appearance of the upper tray body **151**. In some cases, separate chamber walls may be provided to form each upper chamber. In other cases, as shown in FIG. 15, a single chamber wall **153** may be used to define individual chambers within.

(103) As illustrated in FIG. 15, the plurality of upper chambers **152a**, **152b**, and **152c** (as well as fewer or greater number of upper chambers depending on the implementation) may be spaced apart from each other and arranged along the direction A. As explained above with respect to FIG. 13, direction A may be parallel to the rotation axis **cl** of the lower assembly **200**.

(104) As shown in FIG. 14, the upper tray body **151** may include a plurality of upper tray openings **154**, with one opening being provided for each chamber **111**. For example, three upper tray

openings **154** may be defined in an upper surface of the upper tray body **151** to correspond to each of the three chambers **111** underneath. Cold air from the freezer may be guided into the chambers **111** via the openings **154**.

(105) Moreover, the upper ejecting pins **320** of the upper ejector (FIG. 4) may be inserted downward through the upper tray openings **154** to help eject the ice pieces. In some cases, an inlet wall **155** that surrounds and extends upward from a circumference of the upper tray openings **154** may be provided to provide increased structural support.

(106) In some implementations, one or more first connection ribs **155a** may be provided along a circumference of the inlet wall **155** to help prevent the inlet wall **155** from being deformed, for example, when the upper ejector **300** is inserted into the inflow opening **154**. The first connection rib **155a** may connect the inlet wall **155** to the upper tray body **151**. For example, the first connection rib **155a** may be integrated with the circumference of the inlet wall **155** and an outer surface of the upper tray body **151**. In some cases, the plurality of connection ribs **155a** may be disposed along the circumference of the inlet wall **155**.

(107) The two inlet walls **155** corresponding to the second upper chamber **152b** and the third upper chamber **152c** may be connected to each other through the second connection rib **162**. The second connection rib **162** may also help prevent the inlet wall **155** from being deformed.

(108) One of the upper tray openings **154** may be configured as the water receiving hole **112**. For example, as shown in FIG. 14, the water receiving hole **112** may be enlarged and further surrounded by a water supply guide **156** that provides a funnel-like structure for receiving the water supply part **190** (FIG. 4). The water supply guide **156** may be provided as an extension of the inlet wall **155** corresponding to the water receiving chamber, for instance chamber **152b** as illustrated. The water supply guide **156** may be inclined upward and outward from the inlet wall **155**.

(109) The upper tray **150** may further include a first accommodation part **160**. Referring also to FIG. 13, the recess part **122** of the upper case **120** may be accommodated in the first accommodation part **160**. A heater coupling part **124** may be provided in the recess part **122**, and an upper heater **148** (FIG. 20) may be provided in the heater coupling part **124**.

(110) The first accommodation part **160** may be shaped to surround the upper chambers **152a**, **152b**, and **152c**. The first accommodation part **160** may be recessed downward from a top surface of the upper tray body **151**. The heater coupling part **124** to which the upper heater **148** is coupled may be accommodated in the first accommodation part **160**.

(111) The upper tray **150** may further include a second accommodation part **161** that is configured to house the temperature sensor **500** (FIG. 4).

(112) For example, the second accommodation part **161** may be recessed downward from a bottom surface of the first accommodation part **160**. The second accommodation part **161** may be disposed between two adjacent upper chamber. For example, the second accommodation part **161** may be disposed between the first upper chamber **152a** and the second upper chamber **152b**. By providing separate spaces for accommodating the heater and the temperature sensor in this manner, the temperature sensor **500** may be prevented from directly measuring heat coming from the heater **148**. Rather, in the state in which the temperature sensor **500** is accommodated in the second accommodation part **161**, the temperature sensor **500** may contact and measure a temperature of an outer surface of the upper tray body **151**.

(113) Referring to FIGS. 15 and 16, the chamber wall **153** may include a vertical portion **153a** and a curved portion **153b**. The curved portion **153b** is curved outward toward the rotation axis *cl*. As described below with respect to FIG. 30, an outer surface of the curved portion **153b** may help define a water escape passage that is designed to guide excess water out of the chambers **111**. Moreover, the curved surface of the curved portion **153b** can provide a guiding surface for the lower tray **250** when the lower assembly **200** is opened and closed relative to the upper tray **150**.

(114) The upper tray **150** may further include a horizontal extension part **164** that extends horizontally outward from and surrounds the circumference of the upper tray body **151**. The

horizontal extension part **164** may be sandwiched between the upper case **120** and the upper support **170** below to provide a secure coupling of the upper tray **150** to the ice maker **100**.

(115) For example, a bottom surface **164b** of the horizontal extension part **164** may contact the upper support **170**, and a top surface **164a** of the horizontal extension part **164** may contact the upper case **120**. That is, at least a portion of the horizontal extension part **164** may be disposed between the upper case **120** and the upper support **170**.

(116) The horizontal extension part **164** may include a plurality of upper protrusions **165** and **166** that are configured to be inserted into the plurality of upper slots **131** and **132**. In some cases, the protrusion-slot relationship may be reversed.

(117) The plurality of upper protrusions **165** and **166** may include a first upper protrusion **165** and a second upper protrusion **166** disposed at an opposite side of the first upper protrusion **165** with respect to the inflow opening **154**.

(118) The first upper protrusion **165** may be inserted into the first upper slot **131**, and the second upper protrusion **166** may be inserted into the second upper slot **132**. The first upper protrusion **165** and the second upper protrusion **166** may protrude upward from the top surface **164a** of the horizontal extension part **164**. The first upper protrusion **165** and the second upper protrusion **166** may be spaced apart from each other in the direction of the arrow B of FIG. 15. The plurality of first upper protrusions **165** may be arranged to be spaced apart from each other in the direction of the arrow A. In some cases, one or both of the first and second upper protrusion **165**, **166** may have a curved shape.

(119) The upper protrusions **165**, **166** can provide lateral coupling to help restrict a lateral movement and/or deformation of the horizontal extension part **164** relative to the upper case **120** during the ice making and/or the ice ejection process.

(120) The horizontal extension part **164** may further include a plurality of lower protrusions **167** and **168**. The plurality of lower protrusions **167** and **168** may be configured to be inserted into corresponding lower slots that are defined in the upper support **170**. As with the upper protrusions and slots, the protrusion-slot relationship may be reversed.

(121) The plurality of lower protrusions **167** and **168** may include a first lower protrusion **167** and a second lower protrusion **168** disposed at an opposite side of the first lower protrusion **167** with respect to the upper chamber **152**. The first lower protrusion **167** and the second lower protrusion **168** may protrude upward from the bottom surface **164b** of the horizontal extension part **164**.

(122) The first lower protrusion **167** may be disposed opposite the first upper protrusion **165** with respect to the horizontal extension part **164**. The second lower protrusion **168** may be disposed opposite the second upper protrusion **166** with respect to the horizontal extension part **164**. The first lower protrusion **167** may be spaced apart from the vertical wall **153a** of the upper tray body **151**. The second lower protrusion **168** may be spaced apart from the curved wall **153b** of the upper tray body **151**.

(123) Each of the plurality of lower protrusions **167** and **168** may also be provided in a curved shape. Similar to the upper protrusions, the lower protrusions can provide lateral coupling to help restrict a lateral movement and/or deformation of the horizontal extension part **164** relative to the upper support **170** during the ice making and/or the ice ejection process.

(124) In some implementations, the horizontal extension part **164** may include one or more through-holes **169** that may be used, for instance, to receive corresponding coupling bosses of the upper support **170**. One or more of the through-holes **169** may be positioned between adjacent ones of the upper or lower protrusions **165**, **167**. One or more of the through-holes **169** may be positioned between adjacent ones of the upper or lower protrusions **166**, **168**.

(125) Referring to FIGS. 10 and 11, the upper support **170** may include a support plate **171** that is designed to contact and support the upper tray **150**. For example, a top surface of the support plate **171** may contact the bottom surface **164b** of the horizontal extension part **164** of the upper tray **150**. The support plate **171** may define a plate opening **172** through which a portion of the upper

tray body **151** may be inserted through to thereby extend downward from the support plate **171**. The support plate **171** may also include a circumferential wall **174** that surrounds all or a portion of the outer edge of the support plate **171**. Accordingly, the circumferential wall **174** may surround and support an outer side surface of the horizontal extension part **164** of the upper tray **150**. A top surface of the circumferential wall **174** may contact a bottom surface of the upper plate **121** (FIG. **13**).

(126) In some cases, the support plate **171** may include a plurality of lower slots **176** and **177**. The plurality of lower slots **176** and **177** may include a first lower slot **176** into which the first lower protrusion **167** is inserted and a second lower slot **177** into which the second lower protrusion **168** is inserted.

(127) The plurality of first lower slots **176** may be disposed to be spaced apart from each other in the direction of the arrow A on the support plate **171**. Also, the plurality of second lower slots **177** may be disposed to be spaced apart from each other in the direction of the arrow A on the support plate **171**.

(128) The support plate **171** may further include a plurality of coupling bosses **175**. The plurality of coupling bosses **175** may protrude upward from the top surface of the support plate **171**. Each of the coupling bosses **175** may pass through the through-hole **169** of the horizontal extension part **164** and further be inserted into the sleeve **133** (FIG. **12**) of the upper case **120**.

(129) In the state in which the coupling boss **175** is inserted into the sleeve **133** (FIG. **12**), a top surface of the coupling boss **175** may be disposed at the same height as a top surface of the sleeve **133** or disposed at a height lower than that of the top surface of the sleeve **133**.

(130) A coupling member, such as a screw B1 (FIG. **3A**), may be used to couple the upper case **120** to the upper support **170**. The screw B1 may include a body part and a head part having a diameter greater than that of the body part. The screw B1 may be coupled to the coupling boss **175** from an upper side of the coupling boss **175**. When assembled, the head part of the screw B1 may contact and press down on the top surfaces of the sleeve **133** and the coupling boss **175**.

(131) The upper support **170** may further include unit guides **181** and **182** for guiding the connection unit **350** connected to the upper ejector **300**. The unit guides **181** and **182** may, for example, extend upward from opposing side ends of the support plate **171**. The unit guides **181** and **182** may extend upward from the top surface of the support plate **171**. In some cases, the unit guides **181** and **182** may be integral with the circumferential wall **174**.

(132) Each of the unit guides **181** and **182** may include a guide slot **183** that extends along the length of the guides **181**, **182**. Both ends of the ejector body **310** of the ejector **300** may pass outward through each of the guide slots **183** and couple to the connection unit **350**. Accordingly, when the rotation force from the driving unit **180** is transmitted to the ejector body **310** via the connection unit **350**, the ejector body **310** may move vertically up and down along the guide slot **183**.

(133) Referring now to FIGS. **19-21**, the heater coupling part **124**, which can be provided to the upper case **120** to heat the upper tray **150** (FIG. **13**), may include a heater accommodation groove **124a** for accommodating the upper heater **148**. The upper heater **148** may be a wire-type heater. Accordingly, the upper heater **148** may be bendable to correspond to a shape of the heater accommodation groove **124a**.

(134) In some implementations, the heater accommodation groove **124a** may be recessed upward from a bottom surface of the recess part **122** of the upper case **120**. The heater accommodation groove **124a**, and consequently the upper heater **148** accommodated therein, may be arranged to surround an outer perimeter of the opening **123**. Accordingly, the upper heater **148** may be disposed to surround the outer surface of each of the plurality of upper chambers **152** so that the heat from the upper heater **148** may be uniformly transferred to the interior of the plurality of upper chambers **152** of the upper tray **150**. When the upper tray **150** is coupled to the upper case **120**, the heater coupling part **124** may be inserted into the first accommodation part **160** of the upper tray **150** such

that the heater **148** is positioned vertically below the upper tray openings **154**.

(135) In some implementations, as illustrated in FIGS. **19** and **20**, the heater accommodation groove **124a** may be defined between an outer wall **124b** and an inner wall **124c**. In some cases, the upper heater **148** that is accommodated in the heater accommodation groove **124a** may have a diameter that is larger than heater accommodation groove **124a** such that a portion of the upper heater **148** protrudes beyond the heater coupling part **124**. By way of example, a portion of the heater **148** may extend 0.5 mm from the lowermost surface of the heater coupling part **124**.

(136) Accordingly, because the portion of the upper heater **148** protrudes to the outside of the heater accommodation groove **124a** in the state in which the upper heater **148** is accommodated in the heater accommodation groove **124a**, the upper heater **148** may directly contact the upper tray **150**. In some cases, because the heater coupling part **124** is designed to be flush with the contacting surface of the upper tray **150**, the portion of the upper tray **150** that makes contact with the protruded portion of the heater **148** may become deformed to accommodate the heater **148**. In such cases, heat transfer from the heater **148** to the upper tray **150** may be improved.

(137) In some cases, a separation prevention tab **124d** may be provided on one or both of the outer wall **124b** and the inner wall **124c** to prevent the upper heater **148** accommodated in the heater accommodation groove **124a** from being separated from the heater accommodation groove **124a**. The separation prevention tab **124d** may extend from one of the inner wall **124c** and the outer wall **124b** toward the other of the inner wall **124c** and the outer wall **124b**. For example, the tab **124d** may extend to half the distance or less of the separation distance between the inner and outer walls **124c**, **124b** to allow the heater **148** to be inserted into the groove **124a** during assembly but otherwise be prevented from being easily pulled out during use.

(138) As shown in FIG. **20**, the upper heater **148** may include a rounded portion **148c** and a linear portion **148d**. The rounded and linear shapes of the heater **148** may be defined by the corresponding shape provided by the heater accommodation groove **124a**. In some cases, the shapes of the individual heater portions may be pre-defined.

(139) The rounded portions **148c** may be disposed along the circumference of the upper chamber **152** to more effectively transfer heat to the interior of the upper chamber **152**. The linear portions **148d** connect the rounded portions **148c** and help provide heat to portions of the upper tray **150** that are not in contact with the rounded portions **148c**.

(140) As also shown in FIG. **20**, the upper heater **148** may be divided into edge portions **148e** and inner portions **148f**. While FIG. **20** shows a single heating wire that surrounds the entirety of the opening **123**, the edge and inner portions of the upper heater **148** may be provided by shorter heating wires that are connected together. While the illustration depicts one inner portion and two edge portions to correspond to the three upper chambers **152**, a fewer or greater number of inner portions may be provided to correspond to the total number of upper chambers **152** provided.

(141) A length of one edge portion **148e** of the heater **148** may be greater than a length of one inner portion **148f** of the heater **148**. Because the outer upper chamber **152a** or **152c** that corresponds to the edge portion **148e** may have a larger external surface area that is exposed to the cold air in the freezing compartment compared to the inner chamber **152b** (FIG. **15**) that corresponds to the inner portion **148f**, the upper chamber **152a**, **152c** may be cooled more rapidly than the inner chamber **152b**. Accordingly, by providing a longer heating element at the edge portions **148e**, a greater amount of heat may be supplied to the outer chambers **152a**, **152c**, compared to the inner chamber **152b**, thereby helping to equalize the temperature across the chambers.

(142) In some cases, a through-opening **124e** may be defined in a bottom surface of the heater accommodation groove **124a**. When the upper heater **148** is accommodated in the heater accommodation groove **124a**, a portion of the upper heater **148** may be disposed in the through-opening **124e**. For example, the through-opening **124e** may be defined in a portion of the upper heater **148** facing the separation prevention protrusion **124d**. When the upper heater **148** is bent to be horizontally rounded, tension of the upper heater **148** may increase to cause disconnection, and

also, the upper heater **148** may be separated from the heater accommodation groove **124a**. However, by providing the through-opening **124e** in the heater accommodation groove **124a**, a portion of the upper heater **148** may be disposed in the through-opening **124e** to reduce the tension of the upper heater **148**, thereby preventing the heater accommodation groove **124a** from being separated from the upper heater **148**.

(143) As shown in FIG. **21**, a power input terminal **148a** and a power output terminal **148b** of the upper heater **148** may pass upward through a heater through-hole **125** defined in the upper case **120**. The power input terminal **148a** and the power output terminal **148b** passing through the heater through-hole **125** may be connected to one first connector **129a**. A second connector **129c**, which is connected to two wires **129d** that electrically connect to the power input terminal **148a** and the power output terminal **148b**, may be removably coupled to the first connector **129a**.

(144) A first guide part **126** guiding the upper heater **148**, the first connector **129a**, the second connector **129c**, and the wire **129d** may be provided on the upper plate **121** of the upper case **120**. The first guide part **126** may extend upward from the top surface of the upper plate **121** and have an upper end that is bent in the horizontal direction. Thus, the upper bent portion of the first guide part **126** may limit an upward movement of the first connector **126**.

(145) The wires **129d** may be led out to the outside of the upper case **120** after being bent in an approximately “U” shape to prevent interference with the surrounding structures. Since the wire **129d** may include one or more bends, the upper case **120** may further include wire guides **127** and **128** for securing the wires **129d**. The wire guides **127** and **128** may include a first guide **127** and a second guide **128**, which are disposed to be spaced apart from each other in the horizontal direction. The first guide **127** and the second guide **128** may be bent in a direction corresponding to the bending direction of the wire **129d** to minimize damage to the wires **129d**. Thus, each of the first guide **127** and the second guide **128** may include a curved portion.

(146) To limit upward movement of the wire **129d** disposed between the first guide **127** and the second guide **128**, at least one of the first guide **127** and the second guide **128** may include an upper guide **127a** extending toward the other guide.

(147) Referring to FIG. **15**, a cross-sectional view of the upper assembly **110** in which the upper heater **148** is provided to the heater coupling part **124** of the upper case **120** is shown. As illustrated, the upper case **120**, the upper tray **150**, and the upper support **170** are coupled to each other to form the upper assembly **110**. In this state, the first upper protrusion **165** of the upper tray **150** is inserted into the first upper slot **131** of the upper case **120**. Also, the second upper protrusion **166** of the upper tray **150** is inserted into the second upper slot **132** of the upper case **120**. Further, as shown, the first lower protrusion **167** of the upper tray **150** may be inserted into the first lower slot **176** of the upper support **170**, and the second lower protrusion **168** of the upper tray **150** may be inserted into the second lower slot **177** of the upper support **170**.

(148) The coupling boss **175** of the upper support **170** may pass through the through-hole of the upper tray **150** to be accommodated in the sleeve **133** of the upper case **120**. In this state, the screw **B1** (FIG. **3A**) may be coupled to the coupling boss **175** from an upper side of the coupling boss **175**.

(149) When the upper assembly **110** is assembled, the heater coupling part **124** to which the upper heater **148** is coupled may be accommodated in the first accommodation part **160** of the upper tray **150**. In the state in which the heater coupling part **124** is accommodated in the first accommodation part **160**, the upper heater **148** may contact a bottom surface **160a** of the first accommodation part **160**. When the upper heater **148** is accommodated in the heater coupling part **124** having the recessed shape to contact the upper tray body **151**, transfer of heat from the upper heater **148** to the upper tray body **151** may be maximized.

(150) At least a portion of the upper heater **148** may be disposed to vertically overlap the upper chamber **152** to maximize the transfer of heat from the upper heater **148** to the upper chamber **152**. For example, the rounded portion **148c** of the upper heater **148** may vertically overlap the upper

chamber **152**. Thus, a maximum distance between two points of the rounded portion **148c** that are positioned at opposing sides with respect to the upper chamber **152** may be less than a diameter of the upper chamber **152**.

(151) In some implementations, the upper heater **148** may be a DC heater that receives DC power. The upper heater **148** may have a power output of 6 W or less. The upper heater **148** may be a line heater or a heat strip or the like. In some cases, a length of the heater **148** between its input/output terminals may be between 30-40 mm.

(152) The upper heater **148** may be heated to help control the temperature within ice making chambers **111** and in particular the upper chambers **152**. In some cases, the upper heater **148** may be used to temporarily heat the upper chamber **152** to thereby help remove the ice piece during the ice ejection stage. For instance, heat may be added during the ice ejection stage to slightly melt the surface of the ice to thereby promote detachment of the ice piece from the inner surface of the upper chamber **152**.

(153) Referring to FIGS. **23** to **25**, the lower assembly **200** may include a lower tray **250**, a lower support **270**, and a lower case **210**. As illustrated, the lower case **210** may surround and provide support to an upper portion of the lower tray **250**, and the lower support **270** may surround and provide support to a lower portion of the lower tray **250**. The lower case **210**, the lower tray **250**, and the lower support **270** may be coupled to each through various coupling mechanisms as further described below. In some cases, the lower support **270** may be coupled to the connection unit **350**. In some cases, as shown in FIG. **23**, an upper end of the lower case **210** may be coplanar with an upper end of the lower tray **250** when the lower tray **250** is inserted into and coupled to the lower case **210**.

(154) The connection unit **350** may include a first link **352** that receives torque from the driving unit **180** to allow the lower support **270** to rotate together with the first link **352** during the various ice making stages. A second link **356** may be further be connected to the lower support **270** to transfer the rotational motion of the lower support **270** to an up-down movement of the upper ejector **300**.

(155) The first link **352** and the lower support **270** may be connected to each other by an elastic member **360**. For example, the elastic member **360** may be a coil spring. The elastic member **360** may have one end connected to the first link **362** and the other end connected to the lower support **270**. Accordingly, when the first link **362** is rotated by the driving unit **180**, the elastic member **360** may pull up on the lower support **270** to cause the lower support **270** to rotate together with the first link **362**.

(156) The elastic member **360** can provide elastic force to the lower support **270** so that contact between the upper tray **150** and the lower tray **250** may be maintained in the ice making position. For example, referring back to FIG. **6**, the driving unit **180** may over-rotate the lower tray **250** toward the upper tray **150** to ensure that no gaps, which can create seams in the ice, are present between the trays. Such an over-rotation step may be needed because stopping the driving unit **180** immediately upon contact between the upper and lower trays **150**, **250** may still leave some gaps between the two trays. By over-rotating the driving unit **180**, and subsequently the first link **352**, by a small angle, e.g. 1 degree, after the initial contact, the gaps between the two trays may be eliminated. Further, because the lower tray **250** is connected to the first link **352** via the elastic member **360**, the lower tray may stop rotating once the lower tray **250** has been sufficiently compressed toward the upper tray **150** to eliminate any gaps therebetween. Even if the first link **352** continues to be additionally rotated beyond this point, the elastic member **360** can become stretched to thereby prevent the lower tray **250** from also being additionally rotated. Accordingly, additional stresses to the driving unit **180** and other components may be reduced.

(157) In some cases, an overall height of the ice making chamber **111** may be decrease as a result of the over-rotation and subsequent compression between the trays. A stiffness the elastic member **360** may determine the amount of compression. For example, a stiff spring may cause greater

compression compared to a less stiff spring.

(158) As shown in FIG. 23, the first link 352 and the second link 356 may be disposed on both sides of the lower support 270. One or both of the first links 352 on either end may be driven by the driving unit 180. As shown in FIG. 4, the two opposing first links 352 may be connected to each other via a connection shaft 370 that can transmit torque from one link to the other. A hole 358 through which the ejector body 310 and the retaining member 312 of the upper ejector 300 can pass through may be defined in an upper portion of the second link 356.

(159) Referring specifically to FIGS. 24 and 25, the lower case 210 may include a lower plate 211 that is configured to couple to the lower tray 250. For example, an upper surface of the lower tray 250 may contact and become attached to a bottom surface of the lower plate 211.

(160) An opening 212, through which a portion of the lower tray 250 can pass, may be defined in the lower plate 211. For example, when an surface of the lower tray 250 is attached to a bottom surface of the lower plate 211, an upper portion of the lower tray 250 may protrude upward through the opening 212.

(161) The lower case 210 may further include a circumferential wall 214 that extends around a periphery of the opening 212 and that is configured to provide support to the portion of the lower tray 250 that passes upward through the opening 212.

(162) In some implementations, the circumferential wall 214 may include a vertical wall 214a and a curved wall 215. The vertical wall 214a may extend vertically upward from the lower plate 211 to surround a corresponding vertical portion of the upper tray 250. The curved wall 215 also extends generally upward from the lower plate 211 but further includes a curved surface that curves away from the opening 212. The curved portion of the curved wall 215 is designed to support a corresponding curved portion of the upper tray 250.

(163) In some cases, the vertical wall 214a may include a first coupling slit 214b coupled to the lower tray 250. The first coupling slit 214b may be recessed downward from an upper end of the vertical wall 214a. The curved wall 215 may include a second coupling slit 215a that is recessed downward from an upper end of the curved wall 215.

(164) The lower case 210 may further include a first coupling boss 216 and a second coupling boss 217. The first coupling boss 216 may protrude downward from the bottom surface of the lower plate 211. In some cases, a plurality of first coupling bosses 216 may protrude downward from the lower plate 211. The plurality of first coupling bosses 216 may be arranged to be spaced apart from each other in the direction of the arrow A.

(165) The second coupling boss 217 may protrude downward from the bottom surface of the lower plate 211. In some cases, a plurality of second coupling bosses 217 may protrude from the lower plate 211. The plurality of first coupling bosses 217 may be arranged to be spaced apart from each other in the direction of the arrow A.

(166) The first coupling boss 216 and the second coupling boss 217 may be disposed to be spaced apart from each other in the direction of the arrow B. As depicted in FIG. 24, a length of the first coupling boss 216 and a length of the second coupling boss 217 may be different from each other. For example, the first coupling boss 216 may have a length that is shorter than that of the second coupling boss 217.

(167) A first coupling member may be coupled to the first coupling boss 216 at an upper portion of the first coupling boss 216. A second coupling member may be coupled to the second coupling boss 217 at a lower portion of the second coupling boss 217. A groove 215b may be defined in the curved wall 215 to prevent the first coupling member from interfering with the curved wall 215 when the first coupling member is coupled to the first coupling boss 216.

(168) The lower case 210 may include a slot 218 that is configured to allow coupling between the lower case 210 and the lower tray 250. For example, a corresponding portion of the lower tray 250 may be inserted into the slot 218. The slot 218 may be disposed adjacent to the vertical wall 214a.

(169) In some cases, a plurality of slots 218 may be defined to be spaced apart from each other in

the direction of the arrow A. Each of the slots **218** may have a curved shape.

(170) The lower case **210** may further include an accommodation groove **218a** into which a portion of the lower tray **250** is inserted. The accommodation groove **218a** may be defined by recessing a portion of the lower tray **250** toward the curved wall **215**.

(171) The lower case **210** may further include an extension wall **219** for contacting a portion of the circumference of the side surface of the lower plate **211** when it is coupled to the lower tray **250**. The extension wall **219** may extended in a linear direction along the direction of the arrow A.

(172) Referring to FIGS. **26** to **29**, the lower tray **250**, which may be made from a flexible material such as silicone, defines the lower portion of the plurality of ice making chambers **111**, namely the lower chambers **252**. In some cases, the lower tray **250** may be made from a silicone material or other similar material that is more flexible than the material used to make the upper tray **150**.

(173) Accordingly, the lower tray **250** may be restored to its original shape even after being repeatedly deformed during the ice ejection stage to remove the ice pieces from within. Thus, the desired ice shape, for example spherical ice, may be repeatedly formed without substantial variation between ice cycles. Silicone may further be useful due to its ability to withstand extreme temperature variations without deformation.

(174) In one implementation, the lower tray **250** may include a lower tray body **251**, a retaining wall **260**, and a horizontal extension part **254**. The retaining wall **260** may extend generally upward from the top surface of the lower tray body **251**, and the horizontal extension part **254** may extend horizontally outward from an interface between the lower tray body **251** and the retaining wall **260**. The lower tray body **251** defines one or more chambers **252** that forms the lower half of the ice chambers **111**. For example, for spherical ice, the lower chambers **252** may be generally hemispherical in shape. For example, lower chambers **252a**, **252b**, and **252c** shaped for forming spherical ice pieces may be defined within the lower tray body **251**. In particular, the lower chambers may be defined by chamber walls **252d** that are part of the lower tray body **251**.

(175) The lower tray body **251**, the retaining wall **260**, and the horizontal extension part **254** may be provided as a single, integrated piece, for example by being molded together. Accordingly, all three components can be made from the same flexible material. In some cases, a subset of these components may be formed separately and attached together, for example through adhesives or other bonding techniques. For example, the retaining wall **260** and the lower tray body **251** may be formed separately and subsequently attached together, with the horizontal extension part **254** having been formed together with either the retaining wall **260** or the lower tray body **251**. In some cases, the retaining wall **260** and the lower tray body **251** may be formed together as a single piece, with the horizontal extension part **254** being a separate component that is later attached. Different types of materials may be used for the individual components, for example, depending on the particular structural requirements of each.

(176) The lower tray **250** may further include a first extension part **253** between the chamber walls **252d** and the horizontal extension part **254**. The first extension part **253** may be extended along an outer perimeter of the lower tray body **251**.

(177) As explained above with respect to FIG. **5**, the retaining wall **260** of the lower tray **250** extends upward from the lower tray body **251** to help retain an additional volume of water above the lower chambers **252**. In particular, supplied water for filling the upper chambers **152** can be initially held within the retaining wall **260** and later pushed up into the upper chambers **152** based on the closing of the lower tray **250** as explained above with respect to FIGS. **5** to **9**.

(178) In more detail, with reference to FIGS. **26** to **29**, the retaining wall **260** generally extends in a vertically upward orientation from the upper surface **251e** of the lower tray body **251**. An opening defined by the lower edge of the retaining wall **260** may be larger than the opening defined by the chamber walls **252d** at the upper surface **251e** of the lower tray body **251**. Accordingly, a circumferential ledge may be provided around the opening of the lower tray body **251**. When the upper tray **150** and the lower tray **250** are brought together, as shown in FIG. **30**, during the ice

making stage, the bottom surface of the upper tray body **151** makes sealing contact with the circumferential ledge to thereby create fully-formed ice chambers **111** inside the upper and lower tray bodies **151** and **251**. The diameter of the chamber opening defined at the upper surface **251e** of the lower tray body **251** may be equal to the diameter of the chamber opening defined at the lower surface **151e** of the upper tray body **151** such that, when the upper and lower trays are brought together, the interior surfaces of the two tray bodies are flush with each other. In this state, the retaining wall **260** may surround the upper tray body **151** as seen in FIG. **30**.

(179) The retaining wall **260** may include a vertical portion **260a** and a curved portion **260b**. The vertical portion **260a** and the curved portion **260b** of the lower tray's retaining wall **260** are configured to conform to and surround, respectively, the vertical portion **153a** and the curved portion **153b** of the upper tray's chamber wall **153** (FIG. **16**). Thus, the vertical portion **260a** may be extended vertically upward from the lower tray body **251**, and the curved portion **260b** may be curved away from the lower chamber **252** and toward the rotation axis **cl**. The curvature of the curved portion **260b** may be substantially identical to the curvature of the curved portion **153b** such that, when the upper and lower trays are brought together, a water escape passage having a constant thickness may be defined between the outer surfaces of the two curved portions **260b** and **153b**.

(180) The horizontal extension part **254** may extend laterally outward from an interface region between the retaining wall **260** and the lower tray body **251** to define an overall footprint of the lower tray **250**.

(181) The lower tray **250** may include various coupling features to help couple the lower case **210**, the lower tray **250**, and the lower support **270** to each other in a vertically aligned configuration.

(182) For example, the horizontal extension part **254** may include a first upper protrusion **255** that is configured to be inserted into the corresponding slot **218** of the lower case **210**. The first upper protrusion **255** may be formed around the retaining wall **260** in a spaced apart manner and can help restrict a lateral movement and/or deformation of the horizontal extension part **254** relative to the lower case **210** during the ice making and/or the ice ejection process. In some cases, the first upper protrusion **255** may protrude upward from a top surface of the horizontal extension part **254** at a position adjacent to the vertical portion **260a**.

(183) In some implementations, a plurality of first upper protrusions **255** may be arranged to be spaced apart from each other in the direction of the arrow **A**. The first upper protrusion **255** may have a curved shape to increase a length of coupling between the protrusion **255** and the slot **218**.

(184) The horizontal extension part **254** may include a first lower protrusion **257** that is configured to be inserted into a corresponding protrusion groove **287** of the lower support **270** (FIG. **32**). The first lower protrusion **257** may protrude downward from a bottom surface of the horizontal extension part **254**. In some cases, the plurality of first lower protrusions **257** may be arranged to be spaced apart from each other in the direction of arrow **A**.

(185) The first upper protrusion **255** and the first lower protrusion **257** may be positioned opposite to each other with respect to the horizontal extension part **254**. Accordingly, at least a portion of the first upper protrusion **255** may vertically overlap the second lower protrusion **257**.

(186) Many other types of coupling structures may be provided to the lower tray **250**. As another example, a plurality of through-holes **256** may be defined in the horizontal extension part **254**. The plurality of through-holes **256** may include a first through-hole **256a** that is configured to receive the first coupling boss **216** of the lower case **210** and a second through-hole **256b** that is configured to receive the second coupling boss **217** of the lower case **210**.

(187) In some implementations, the plurality of through-holes **256a** may be spaced apart from each other in the direction of the arrow **A** (FIG. **26**). Similarly, the plurality of second through-holes **256b** may be spaced apart from each other in the direction of the arrow **A**. In some cases, the plurality of first through-holes **256a** and the plurality of second through-holes **256b** may be disposed at opposite sides of the horizontal extension part **254** with respect to the lower chamber **252**.

(188) A portion of the plurality of second through-holes **256b** may be positioned between adjacent ones of the first upper protrusions **255**. Also, a portion of the plurality of second through-holes **256b** may be positioned between adjacent ones of the first lower protrusions **257**.

(189) The horizontal extension part **254** may also include one or more second upper protrusions **258** (FIG. **29**) that are positioned opposite the first upper protrusions **255** with respect to the lower chamber **252**.

(190) In some cases, the second upper protrusion **258** may be formed to extend alongside the curved portion **260b** in a spaced apart manner and can help restrict a lateral movement and/or deformation of the horizontal extension part **254** relative to the lower case **210**. The second upper protrusion **258** may protrude upward from a top surface of the horizontal extension part **254** at a position adjacent to the curved portion **260b**. In some cases, the plurality of second upper protrusions **258** may be arranged to be spaced apart from each other in the direction of the arrow A (FIG. **26**). The second upper protrusion **258** may be accommodated in the corresponding accommodation groove **218a** of the lower case **210** (FIG. **25**). In some cases, when the lower tray **250** is coupled to the lower case **210**, the second upper protrusions **258** may be accommodated within the curved wall **215** of the lower case **210** (FIG. **24**).

(191) In some implementations, the retaining wall **260** of the lower tray **250** may include one or more first coupling protrusions **262** that are configured to couple the retaining wall **260** to the lower case **210**. In some cases, each of the first coupling protrusions **262** may be button-like structures that protrude laterally from the vertical portion **260a** of the retaining wall **260**. In particular, the first coupling protrusion **262** may be disposed on an upper portion of an outward facing surface of the vertical portion **260a**.

(192) The first coupling protrusion **262** may include a neck part **262a** having a smaller diameter compared to the remaining portion of the protrusion **262**. In use, the neck part **262a** may be inserted into a first coupling slit **214b** that is defined in the circumferential wall **214** of the lower case **210** to couple the retaining wall **260** to the lower case **210**. Once secured, a portion of the circumferential wall **214** may be positioned between an inner surface of the first coupling protrusion **262** and an outer surface of the vertical portion **260a**. In some cases, the uppermost portion of the first coupling protrusion **262** may be coplanar with the uppermost edge of the vertical portion **260a** of the retaining wall **260**.

(193) In some cases, as shown in FIGS. **28** and **29**, the retaining wall **260** may further include one or more second coupling protrusions **260c** that are configured to couple the retaining wall **260** to the lower case **210**.

(194) The second coupling protrusion **260c** may protrude laterally from the curved portion **260b** of the retaining wall **260** and be configured to be inserted into a corresponding second coupling slit **215a** that is defined in the circumferential wall **214** of the lower case **210**. By providing coupling between the curved portion **260b** of the lower tray **250** and the circumferential wall **214** of the lower case **210**, the curved shape of the curved portion **260b** may be maintained during rotation of the lower assembly **200**. Alternatively, or additionally, the curved portion **260b** of the lower tray **250** may be made thicker compared to the remaining portions of the retaining wall **260** for increased stiffness.

(195) In some implementations, the horizontal extension part **254** may include a second lower protrusion **266**. The second lower protrusion **266** may be disposed at an opposite side of the second lower protrusion **257** with respect to the lower chamber **252**. The second lower protrusion **266** may protrude downward from a bottom surface of the horizontal extension part **254** and be linearly extended along an outer edge of the horizontal extension part **254**. One or more of the plurality of first through-holes **256a** may be defined between the second lower protrusion **266** and the lower chamber **252**. When the lower tray **250** is coupled to the lower support **270**, the second lower protrusion **266** may be accommodated within a corresponding guide groove that is defined in the lower support **270** (FIG. **32**).

(196) In some cases, the horizontal extension part **254** may further a side restriction part **264**. The side restriction part **264** may be configured to restrict a horizontal movement of the lower tray **250** when it is coupled to the lower case **210** and the lower support **270**.

(197) The side restriction part **264** may protrude laterally from the horizontal extension part **254** and can have a vertical length greater than a thickness of the horizontal extension part **254**. Thus, an upper portion of the side restriction part **264** may contact a side surface of the lower case **210**, and its lower portion may contact a side surface of the lower support **270**.

(198) Referring to FIGS. **30** and **31**, when the upper tray **150** and the lower tray **250** are brought together, for example during the ice making stage, the upper chamber **152** and the lower chamber **252** come in contact with each other to form a complete ice chamber **111** that defines, for instance, a spherical shape of the ice piece to be generated. Specifically, the bottom surface **151a** of the upper tray body **151** contacts the upper surface **251e** of the lower tray body **251**. As described above, when the upper and lower trays are brought together in this manner, additional elastic force may be applied by the elastic member **360** to further compress the two tray bodies toward each other, thereby helping to eliminate gaps between the two tray bodies.

(199) When the lower assembly **200** and the upper assembly **110** are brought together as shown in FIG. **30**, a first water escape passage **261a** may be defined between a vertical portion **153a** of the upper tray **150** and a vertical portion **260a** of the lower tray **250**. Similarly, a second water escape passage **261b** may be formed between a curved portion **153b** of the upper tray **150** and a curved portion **260b** of the lower tray **250**.

(200) The first water escape passage **261a** may be formed by configuring an outer surface of the vertical portion **153a** to be spaced apart from an inner surface of the vertical portion **260a** when the retaining wall **260** surrounds the chamber wall **153** (e.g. in the ice making stage). The second water escape passage **261b** may be formed by configuring an outer surface of the curved portion **153b** of the upper tray **150** to be spaced apart from an inner surface of the curved portion **260b** of the lower tray **250** when the retaining wall **260** surrounds the chamber wall **153** (e.g. in the ice making stage).

(201) By way of example, the first and water escape passages **261a**, **261b** may be between 1 to 2 mm in thickness. In some cases, the first and water escape passages **261a**, **261b** may have a thickness of less than 1 mm. In some cases, the thickness may be less than 0.5 mm.

(202) Referring also to FIGS. **5** to **9**, when the water retained within the retaining wall **260** is pushed up to fill the upper chambers **252** by the upward rotation of the lower assembly **200**, excess water may flow into the water escape passages **261a**, **261b**. That is, excess water may be guided away from the ice making chamber **111** through the water escape passages **261a**, **261b** instead of overflowing through the upper tray openings **154**. Excess water in the water escape passage **261a**, **261b** may flow out into the freezer. Alternatively, or additionally, thin pieces of ice that form within the water escape passages **261a**, **261b** may break up and fall out based on the back and forth movement of the lower assembly **200**.

(203) In some cases, the uppermost portion of the retention wall **260** may be positioned vertically higher than the upper tray openings **154**.

(204) With reference to FIGS. **29** to **31**, the lower portion of the lower tray body **251** includes a stepped portion **251a** and a deformable portion **251b**. In some cases, the stepped portion **251a** may surround a circumference of the deformable portion **251b**.

(205) The stepped portion **251a** may be in a ring shape and is protruded downward from the lower tray body **251**. A lower surface of the stepped portion **251a** may be flat and can provide a heater contact surface for a lower heater **296** (FIG. **36**). The stepped portion **251a** may be positioned at any height around the circumference of the lower tray body **251**. In some implementations, in order to provide heat to a lower portion of the chamber **252** during the ice making process, the stepped portion **251a** may be positioned at a height that is below the halfway point of the height of the lower chamber **252**. In some cases, the stepped portion **251a** may be positioned at the lowermost portion of the lower tray body **251**. In some cases, as illustrated in FIG. **29**, only the deformable

portion **251b** of the lower tray body **251** may be positioned below the stepped portion **251a**. An inner diameter of the stepped portion **251a** may be larger than a diameter of the ejector pin **320** such that the ejector pin **320** can pass through the stepped portion **251a** during the ice ejection stage.

(206) The deformable portion **251b** may change from a first shape to a second shape during the ice generation process. For example, as shown in FIG. **30**, the deformable portion **251b** may have a convex shape (i.e. first shape) before ice is formed within the ice chamber **111**; however, after the ice **I** is formed within the ice chamber **111**, outward expansion of the ice **I** may exert an outward force on the deformable portion **251b** to change the convex shape to a concave shape (i.e. second shape).

(207) A recess part **251c** may be defined at a lower surface of the deformable portion **251b** to allow the deformable portion **251b** to more readily transition from the first shape to the second shape. For example, due to the presence of the recess part **251c**, the deformable portion **251b** may have a uniform thickness across its entire before and after the shape change. In some cases, the recess part **251c** may reduce a thickness of the deformable portion **251b** relative to the remaining portions of the lower tray body **251** to thereby increase flexibility of the deformable portion **251b**. Accordingly, the deformable portion **251b** may be able to more easily transition between the first and second shapes. By adjusting the thickness or, in some cases, the material properties of the deformable portion **251b**, the amount of expansion force required to transition from the first shape to the second shape may be adjusted.

(208) By including an appropriately designed deformable portion **251b** to the lower chamber **252**, the desired final shape of the ice generated within the ice chamber **111** may be achieved. Notably, because water expands when phase-changed into solid ice, the shape of the ice chamber **111** itself may change as the water expands and turns into ice. For instance, a spherical chamber into which water is supplied may expand and become distorted when the water contained inside freezes. This is especially true in ice maker configurations in which the top portion of the chamber may be colder than the bottom portion of the chamber, thus causing the water to freeze starting from the top and moving down (see FIG. **42**). In such cases, the expansion/distortion of the ice chamber **111**, which is made of a flexible material, may largely be localized to the lowermost portion of the chamber that freezes last. Consequently, the lowermost portion of the ice formed inside such a chamber may include a nipple-like protrusion.

(209) In contrast, by including the deformable portion **251b** at the lowermost portion of the chamber **111**, the anticipated expansion of the ice in that region can be accounted for. For example, by including a convex deformable portion at the lower part of the lower chamber **252**, a localized expansion of ice in that region can cause the convex portion to become concave, thus transforming the shape of the lower chamber **252** to be closer to the desired hemispherical shape. In turn, a more hemispherical lower portion of the ice can lead to a more spherical shape overall.

(210) The shape and location of the deformable portion **251b** may be adjusted depending on the specific location and size of the expected region of expansion/deformation.

(211) Referring now to FIGS. **32** to **34**, the lower support **270** of the lower assembly **20** may include a support body **271** that is configured to provide support to the lower tray **250**. In particular, the support body **281** may define three chamber accommodation portions **272** that are configured to surround and provide support to corresponding chamber walls **252d** of the lower tray body **251**. For example, if the lower tray body **251** has a generally hemispherical shape that is defined by the chamber walls **252d**, then the chamber accommodation portion **272** may be shaped correspondingly to have a hemispherical shape. Accordingly, the lower support **270** can help prevent an outward expansion of the lower tray body **251**, for example during ice generation when outward expansion forces can act on the lower tray body **251**. The lower support **270** can be made from plastic or other similar materials that may be more rigid than the lower tray body **251**.

(212) The support body **271** may define one or more openings **274** through which the lower ejector

400 can pass during the ice ejection stage. For example, three lower openings **274** may be defined to correspond to the three chamber accommodation parts **272** in the support body **271**. Referring also to FIGS. **30** and **31**, the lower openings **274** can provide space through which the deformable portion **251b** of the lower tray body **251** can expand outward. That is, while the remaining portions of the chamber accommodation portion **272** serve to constrain the contacted portions of the lower tray body **251** from expanding outward, the lower opening **274** may overlap with the deformable portion **251b** to allow a change from the first shape (e.g., convex shape) to a second shape (e.g., concave shape). Accordingly, as shown in FIG. **30**, a diameter **D1** of the deformable portion **251b** may be less than a diameter **D2** of the lower opening **274**.

(213) In some implementations, a reinforcement rib **275** may be provided around a circumference of the lower opening **274** to provide additional structural reinforcement. Structural reinforcement may also be provided through one or more connection ribs **273** that are provided across adjacent ones of the chamber walls **252d**. The lower support **270** may also include a stepped portion **285** that extends laterally from an upper end of the support body **271**.

(214) In some cases, the lower support may include a second extension wall **286** that is stepped and extends from an edge of the stepped portion **285**. Thus, a top surface of the second extension wall **286** may be positioned vertically higher than the stepped portion **285**.

(215) The first extension part **253** of the lower tray **250** (FIG. **30**) may be seated on a top surface **271a** of the support body **271**, and the second extension wall **286** may surround the side surface of the first extension part **253** of the lower tray **250**. Here, the second extension wall **286** may contact the side surface of the first extension part **253** of the lower tray **250**.

(216) The lower support **270** may further include protrusion grooves **287** that is configured to receive and secure the first lower protrusion **257** of the lower tray **250**. Each of the protrusion grooves **287** may have a matching curved shape. The protrusion groove **287** may be defined in the second extension wall **286**.

(217) The lower support **270** may further include one or more first coupling grooves **286a** to which a first coupling member **B2** (FIG. **34**), which is passed through the first coupling boss **216** of the upper case **210**, can be coupled. In some cases, the one or more first coupling grooves **286a** may be defined in the second extension wall **286**.

(218) The plurality of first coupling grooves **286a** may be arranged to be spaced apart from each other in the direction of the arrow **A** on the second extension wall **286**. A portion of the plurality of first coupling grooves **286a** may be defined between adjacent ones of the protrusion grooves **287**.

(219) In some cases, the lower support **270** may define a boss through-hole **286b** through which the second coupling boss **217** of the upper case **210** can pass. The boss through-hole **286b** may be provided, for example, in the second extension wall **286**. A sleeve **286c** that surrounds the second coupling boss **217**, which has passed through the boss through-hole **286b**, may be disposed on the second extension wall **286**. The sleeve **286c** may have a cylindrical shape with an open lower end. A second coupling member **B3** may be coupled to the second coupling boss **217** from a lower side of the lower support **270**.

(220) The sleeve **286c** may have a lower end that is disposed at the same height as a lower end of the second coupling boss **217**. Alternatively, the lower end of the sleeve **286c** may be disposed at a height lower than that of the lower end of the second coupling boss **217**. Accordingly, when the second coupling member **B3** is provided, the head part of the second coupling member **B3** may contact bottom surfaces of the second coupling boss **217** and the sleeve **286c**. Alternatively, the head part may contact a bottom surface of the sleeve **286c**.

(221) The lower support **270** may further include an outer wall **280** that surrounds the lower tray body **251**. The outer wall **280** may be extended downward from an outer perimeter of the second extension wall **286**. The lower support **270** may further include a plurality of hinge bodies **281** and **282** that are configured accommodate, respectively, hinge supports **135** and **136** of the upper case **210**. The plurality of hinge bodies **281** and **282** may be spaced apart from each other in a direction

of the arrow A (FIG. 32). Each of the hinge bodies **281** and **282** may define therein a second hinge hole **281a**. The shaft connection part **353** of the first link **352** may pass through the second hinge hole **281**. The connection shaft **370** may be connected to the shaft connection part **353**.

(222) A distance between the plurality of hinge bodies **281** and **282** may be less than a distance between the plurality of hinge supports **135** and **136**. Thus, the plurality of hinge bodies **281** and **282** may be disposed between the plurality of hinge supports **135** and **136**.

(223) The lower support **270** may further include a coupling shaft **283** to which the second link **356** is rotatably coupled. The coupling shaft **383** may be disposed on each of both surfaces of the outer wall **280**.

(224) In some cases, the lower support **270** may include an elastic member coupling part **284** to which the elastic member **360** is coupled. The elastic member coupling part **284** may define a space in which a portion of the elastic member **360** is accommodated. The elastic member coupling part **284** may include a hook part **284a** to which a lower end of the elastic member **360** can be hooked.

(225) Referring to FIGS. 35 to 37, the lower heater **296** may be provided to the lower support **270** in order to provide heat to the lower tray **250** during the ice making process. In particular, the heater **296** may provide heat to the lower chamber **252** during the ice making process to cause the ice within the ice chamber **111** to start freezing from the upper side of the chamber **111**.

Accordingly, by controlling the propagation of ice formation in this manner, air bubbles within the ice, which can give rise to hazy/opaque ice, may be directed to the bottommost portion of the ice. Thus, a substantial portion of the ice made within the chamber **111** may be transparent. Similar to the upper heater **148**, the lower heater **296** may be a flexible wire-type heater, for example a line heater or a heat strip.

(226) The lower heater **296** may be installed on the lower support **270** to make contact with and heat the lower tray **250**. For example, the lower heater **296** may contact the lower tray body **251** to thereby provide heat to the lower chamber **252**. In particular, the lower heater **296** may be disposed around a circumference of the chamber walls **252d**.

(227) The lower support **270** may further include a heater coupling part **290** to which the lower heater **296** is coupled. The heater coupling part **290** may include a heater accommodation groove **291** that is recessed from the chamber accommodation part **272** of the lower tray body **251**. The heater coupling part **290** may thus include an inner wall **291a** and an outer wall **291b**. In some cases, the inner wall **291a** may have a ring shape, and the outer wall **291b** may surround the inner wall **291a**. When the lower heater **296** is accommodated in the heater accommodation groove **291**, the lower heater **296** may surround at least a portion of the inner wall **291a**.

(228) The lower support **270** may define lower openings **274**. The lower opening **274** may be defined in a region defined by the inner wall **291a**. Thus, when the chamber wall **252d** of the lower tray **250** is accommodated in the chamber accommodation part **272**, the chamber wall **252d** may contact a top surface of the inner wall **291a**. The top surface of the inner wall **291a** may be a rounded surface corresponding to the chamber wall **252d** having the hemispherical shape.

(229) The lower heater may have a diameter greater than a recessed depth of the heater accommodation groove **291** such that a portion of the lower heater **296** protrudes to the outside of the heater accommodation groove **291** in the state in which the lower heater **296** is accommodated in the heater accommodation groove **291**. The protruded portion of the lower heater **296** may be pressed into the lower tray body **251** to allow for better heat transfer into the lower tray body **251**. In some cases, the lower heater **296** may protrude approximately 0.5 mm above the accommodation groove **291**.

(230) In some implementations, a separation prevention protrusion **291c** may be provided on one or both the outer wall **291b** and the inner wall **291a** to help prevent the lower heater **296** accommodated in the heater accommodation groove **291** from being separated from the heater accommodation groove **291**.

(231) The lower heater **296** may be accommodated in the heater accommodation groove **291** from

an upper side of the outer wall **291a** toward the inner wall **291a**. Thus, the separation prevention protrusion **291c** may be disposed on the inner wall **291a** to prevent the lower heater **296** from interfering with the separation prevention protrusion **291c** while the lower heater **296** is accommodated in the heater accommodation groove **291**. The separation prevention protrusion **291c** may protrude from an upper end of the inner wall **291a** toward the outer wall **291b**.

(232) In some cases, the separation prevention protrusion **291c** may extend to half the distance or less of the separation distance between the inner and outer walls **291a**, **291b** to allow the heater **296** to be inserted into the groove **291** during assembly but otherwise be prevented from being easily pulled out during use.

(233) As illustrated in FIG. **36**, when the lower heater **296** is accommodated in the heater accommodation groove **291**, the lower heater **296** may be classified into a rounded portion **296a** and a linear portion **296b**. For example, the lower heater **296** may be divided into the rounded portion **296a** and the linear portion **296b** to correspond to the rounded portion and the linear portion of the heater accommodation groove **296**. The rounded portion **296a** may be disposed along the circumference of the lower chamber **252**. The linear portion **296b** may be used to connect the rounded portions **296a** to each other.

(234) As seen in FIG. **35**, a through-opening **291d** may be defined at a bottom surface of the heater accommodation groove **291**. Thus, when the lower heater **296** is accommodated in the heater accommodation groove **291**, a portion of the lower heater **296** may be accommodated in the through-opening **291d**. For example, the through-opening **291d** may be defined in a portion of the lower heater **296** facing the separation prevention protrusion **291c**.

(235) When the lower heater **296** is bent, increased tension may be applied to the lower heater **296**, thus causing the heater from being disconnected and/or separated from the heater accommodation groove **291**. However, a portion of the lower heater **296** may be disposed in the through-opening **291d** to reduce tension on the lower heater **296**, thereby preventing the heater accommodation groove **291** from being separated from the lower heater **296**.

(236) The lower support **270** may include a first guide groove **293** that guides a power input terminal **296c** and a power output terminal of the lower heater **296** accommodated in the heater accommodation groove **291**. The lower support **270** may also include a second guide groove **294** that extends in a transverse direction to the first guide groove **293**. For example, the first guide groove **293** may extend in a direction of an arrow B (FIG. **36**) in the heater accommodation part **291**.

(237) In some cases, the second guide groove **294** may extend from an end of the first guide groove **293** in a direction of an arrow A (FIG. **36**). In some cases, the direction of the arrow A may be parallel to the rotational central axis C1.

(238) In some implementations, as seen in FIG. **36**, the first guide groove **293** may extend from one of the left and right chamber accommodation. For example, the first guide groove **293** may extend from the leftmost chamber accommodation part among the three chamber accommodation parts.

(239) In some implementations, the power input terminal **296c** and the power output terminal **296d** of the lower heater **296** may be connected to a first connector **297a**. Additionally, a second connector **297b** to which two wires **298** corresponding to the power input terminal **296c** and the power output terminal **296d** are connected may be connected to the first connector **297a**. When the first connector **297a** and the second connector **297b** are connected to each other, the first connector **297a** and the second connector **297b** may be accommodated in the second guide groove **294**.

(240) The wire **298** connected to the second connector **297b** may be led out from the end of the second guide groove **294** to the outside of the lower support **270** through an lead-out slot **295** defined in the lower support **270**.

(241) In some cases, different amount of heat may need to be provided to the individual lower chambers **252** to achieve a uniform temperature across the multiple chambers. For example, because the outer chambers may be exposed to more cold air than the middle chambers, more heat

may need to be provided to the outer chambers to achieve uniform temperature across all the chambers. As another example, because some heat may be generated by the power input terminal **296c** and the power output terminal **296d**, a chamber that is closest to these terminals, for example, may experience an increased temperature compared to the remaining chambers. Non-uniform heat provided across the chambers may lead to different levels of transparency for the ice generated within those chambers.

(242) Accordingly, in some implementations, additional heater grooves **292** may be provided around the chamber accommodation portion **272** to help achieve uniform heat distribution. For example, as seen in FIGS. **35** and **36**, the additional heater groove **292** may extend outward from the main heater accommodation groove **291**. Accordingly, because a contact area between the chamber accommodation part **272** and the lower heater **296** may increase in the region of the additional heater groove **292**, the amount of heat provided to that region may correspondingly increase. That is, the additional heater groove **292** helps provide a heater extension part **296e** for providing additional heat to a specific region of the lower tray body **251**.

(243) In some cases, a protrusion **292a** may be provided in conjunction with the additional heater groove **292** to help secure the heater extension part **296e**. While the implementation shown in FIG. **36** showed one possible location of the additional heater groove **292** and the corresponding heater extension part **296e**, the heater extension part **296e** may be similarly provided to other locations around the lower tray **251** as needed. The upper heater **148**, as seen in FIG. **20**, may be similarly configured to provide additional heating to different portions of the upper tray **150**.

(244) In some cases, as seen in FIG. **37**, the wire **298** that is led out of the lower support **270** may pass through a wire through-slot **138** defined in the upper case **120** to extend upward from the upper case **120**. A restriction guide **139** that is configured to restrict the movement of the wire **298** passing through the wire through-slot **138** may be provided in the wire through-slot **138**. The restriction guide **139** may include several bends to thereby confine the wire **298** within the restriction guide **139**.

(245) Referring to FIG. **38**, the refrigerator may include a control unit **700** for controlling the upper heater **148** and the lower heater **296**. For example, the control unit **700** may adjust an output of the lower heater **296** during the ice making process.

(246) Referring to FIG. **39**, an example process flow for generating ice using the ice maker **100** is shown.

(247) Initially, the lower assembly **200** may move to a water supply position (S1). As explained above with respect to FIG. **5**, top surface **251e** of the lower tray **250** may be spaced apart from the bottom surface **151e** of the upper tray **150**. The driving unit **180** may have rotated the lower assembly **200** in either direction to arrive at this stage. In some cases, the bottom surface **151e** of the upper tray **150** may be disposed at a height that is equal to that of the rotation axis C1 of the lower assembly **200**.

(248) In this state, the angle between the top surface **251e** of the lower tray **250** and the bottom surface **151e** of the upper tray **150** at the water supply standby position of the lower assembly **200** may be approximately 8 degrees.

(249) The supplying of water may be started in (S2). For example, water flows to the water supply part **190** through a water supply tube connected to an external water supply source or a water tank of the refrigerator **1**. Subsequently, the water is guided by the water supply part **190** and supplied to the ice chamber **111**. Here, the water is supplied to the ice chamber **111** through one of the upper tray openings **154**, namely water receiving hole **112**, of the upper tray **150**.

(250) As described above, since the top surface **251e** of the lower tray **250** and the bottom surface **151e** of the upper tray **150** are spaced apart from each other at this state, water that is supplied to just one of the chambers may overflow and flow into the remaining chambers as well.

(251) Thus, the water may be fully filled in each of the plurality of lower chambers **252** of the lower tray **250**.

(252) Upon completion of the water supply stage, the lower assembly **200** is rotated toward the upper assembly **110** to the ice making position (S3). Due to this upward movement of the lower assembly **200**, additional volume of water contained by the retaining wall **260** is directed into the upper chambers **152**. An over-rotation of the driving unit **180** may take place at this stage to further press the lower tray **250** into the upper tray **150**, thereby helping to eliminate gaps between the two trays.

(253) Water within the chambers is allowed to freeze during the ice making process (S4).

(254) After the ice making is started, the control unit **700** determines whether a turn-on condition of the lower heater **296** is satisfied (S5). That is, by way of example, the lower heater **296** may be turned on only when the turn-on condition of the lower heater **296** is satisfied.

(255) Specifically, the lower heater **296** may not be turned on until the water starts to phase-change into ice. Otherwise, if the lower heater **296** is turned on before reaching the freezing point of the water in the ice chamber **111**, a rate at which the temperature of the water reaches the freezing point may be lowered by the heat of the lower heater **296**, resulting in a reduced ice making rate.

(256) The control unit **700** may determine when the turn-on condition of the lower heater **296** is satisfied by determining when a temperature detected by the temperature sensor **500** reaches a turn-on reference temperature. For example, the turn-on reference temperature may be a temperature at which the freezing of water starts at the uppermost side (an inflow opening side) of the ice chamber **111**.

(257) In this implementation, since the ice chamber **111** is blocked by the upper tray **150** and the lower tray **250** except for the inflow opening **154**, the water in the ice chamber **111** may directly contact the cold air through the inflow opening **154** to make ice from the uppermost side in which the inflow opening is disposed in the ice chamber **111**.

(258) When water is frozen in the ice chamber **111**, a temperature of the ice in the ice chamber **111** may be below zero. Also, the temperature of the upper tray **150** may be higher than that of the ice in the ice chamber **111**.

(259) In some implementations, the temperature sensor **500** may detect the temperature of the upper tray **150** by contacting the upper tray **150** without directly detecting the temperature of the ice. According to the above-described arranged structure, to determine that making of ice is started in the ice chamber **111** on the basis of the temperature detected by the temperature sensor **500**, the turn-on reference temperature may be set to the below-zero temperature.

(260) That is, when the temperature detected by the temperature sensor **500** reaches the turn-on reference temperature, which is below zero, and the temperature of the ice in the ice chamber **111** is lower than the turn-on reference temperature, it may be indirectly determined that the ice has formed in the ice chamber **111**.

(261) When the lower heater **296** is turned on, heat of the lower heater **296** is transferred to the lower tray **250** (S6).

(262) Thus, when the ice making is performed in the state where the lower heater **296** is turned on, ice may be made from the upper side in the ice chamber **111** because the heat is supplied to the lower chamber **252** through the water contained in the lower chamber **252**.

(263) When the ice starts to form from the upper side of the ice chamber **111**, the bubbles in the ice chamber **111** may move downward. That is, because a density of water is greater than that of ice, the bubbles in the water may easily move downward to be gathered downward.

(264) When the ice chamber **111** has a spherical shape, the horizontal cross-sectional area for each height of the ice chambers **111** are different from each other. Then, assuming that the same amount of cold air is supplied to the ice chamber **111**, if the output of the lower heater **296** is the same, the horizontal cross-sectional area for each height of the ice chambers **111** may be different from each other, and thus, ice may be made at heights different from each other. That is to say, the height at which ice is made per unit time may be non-uniform. In this case, the bubbles in the water may not be properly moved downward and instead become trapped in the ice so that the ice becomes

opaque.

(265) Accordingly, the control unit **700** may control the output of the lower heater **296** according to the height of the ice made in the ice chamber **111** (S7).

(266) In particular, the horizontal cross-sectional area of the ice increases from the upper side to the lower side of the upper chamber **152**, is maximized at a boundary between the upper tray **150** and the lower tray **250**, and decreases again to the lower side of the lower chamber **252**. The control unit **700** may thus allow the output of the lower heater **296** to vary in response to a variation in horizontal cross-sectional area according to the height.

(267) The control unit **700** may determine whether the ice making is completed based on the temperature sensed by the temperature sensor **500** (S8). When it is determined that the ice making is completed, the control unit **700** may turn off the lower heater **296** (S9).

(268) In some implementations, the distance between the temperature sensor **500** and each of the ice chambers **111** may be different from each other. Thus, to determine that the making of ice is completed in all the ice chambers **111**, ice ejection may be started after a certain time elapses from a time point at which it is determined that the ice making is completed.

(269) When the ice making is completed, to eject the ice, the control unit **700** may operate the upper heater **148** (S10).

(270) When the upper heater **148** is turned on, the heat of the upper heater **148** is transferred to the upper tray **150**, and thus, the ice may be separated from the surface (the inner surface) of the upper tray **150**. The heat of the upper heater **148** may also be transferred to the contact surface between the upper tray **150** and the lower tray **250** to help separate the bottom surface **151a** of the upper tray **150** and the top surface **251e** of the lower tray **250** from each other.

(271) After the upper heater **148** has operated for a set time, the control unit **700** may turn off the upper heater **148**. Also, the driving unit **180** may be operated at this time so that the lower assembly **200** is rotated away from the upper assembly **110** to the ice ejection position (S11).

(272) Referring to FIGS. **40A**, **40B**, **41**, and **42**, the controlled variation of the power output of the lower heater **296** in response to variations in the horizontal cross-sections of the ice piece is illustrated.

(273) In particular, when the ice chamber is divided into the reference intervals, as shown in FIG. **40A**, the heights of each of the sections A to H may be the same. Because of the deformable portion **251b** at the bottom of the ice chamber, the height of the section I may be less than the other sections. Alternatively, all the divided sections may have the same height.

(274) In the example of FIG. **40A**, since section E has the largest diameter, it represents the maximum section volume. Thus, assuming generally uniform cooling conditions, the ice making rate in section E may be the slowest, with the rates in the smallest sections A and I being the fastest. Due to the varying ice making rates across the sections, transparency of the ice in each section—which is dictated by the presence of trapped air bubbles—may also vary across the sections. Some sections, for example, may freeze too quickly before allowing the air bubbles to escape.

(275) By controlling the output of the lower heater **296**, the freezing rate and direction may be controlled such that the air bubbles move downward toward the lowermost portion of the ice chamber **111** during the ice making process.

(276) For example, as shown in FIG. **40B**, an output W5 of the lower heater **296** corresponding to the section E may be set to a minimum value to maximize the amount of cooling to that relatively large region.

(277) Because the relatively smaller volume of water in section D may freeze quicker than section E, air bubbles may become trapped in section D. Accordingly, in order to delay the ice making rate in section D, a corresponding output W4 may be set to a value greater than the output W5 of the lower heater **296** in the section E. Thus, section D may be prevented from becoming frozen before section E.

(278) By the same rationale, output W3 corresponding to section C, output W2 according to section

B, and output W1 corresponding to section A may be increasingly greater.

(279) To prevent the water in section F from freezing before section E, which would cause air bubbles in section E to become trapped, an output W6 of the lower heater 296 that corresponds to Section F may be greater than output W5. Similarly, output W7 may be greater than output W6, and output W8 may be greater still than output W7. Output W9 corresponding to section I, which has the smallest volume of water and thus susceptible to freezing the quickest, can thus be the largest.

(280) Further referring to FIG. 42, by adjusting the power output of the lower heater 296 in the manner described above, water W within the chamber 111 can be made to freeze starting at the top such that ice I first forms at the top of the chamber and then gradually propagates toward the bottom, in the process driving the air bubbles downward.

(281) Referring to FIG. 41, the temperature detected by the temperature sensor 500 may generally decrease as a greater portion of the ice chamber 111 freezes. By storing such temperature patterns in a memory, the controller 700 can use these temperatures as reference temperatures to help determine the progress of ice propagation and to apply the corresponding amount of heat.

(282) For example, when the temperature detected by the temperature sensor 500 reaches the reference temperature of the next section in the present section, the control unit 700 adjusts an output of the lower heater 296 corresponding to the present section to match to an output corresponding to the next section.

(283) Although implementations have been described with reference to a number of illustrative implementations thereof, it should be understood that numerous other modifications and implementations can be devised by those skilled in the art that will fall within the spirit and scope of the principles of this disclosure. More particularly, various variations and modifications are possible in the component parts and/or arrangements of the subject combination arrangement within the scope of the disclosure, the drawings and the appended claims. In addition to variations and modifications in the component parts and/or arrangements, alternative uses will also be apparent to those skilled in the art.

Claims

1. An ice maker comprising: a first tray that defines a first chamber, the first tray comprising a chamber wall defining the first chamber; a second tray that defines a second chamber, the second tray comprising (i) a tray body defining the second chamber recessed in a first direction, (ii) a tray circumferential wall extending from the tray body in a second direction opposite to the first direction, and (iii) a first coupling protrusion that protrudes from an outer surface of the tray circumferential wall; and a case configured to support the second tray, the case comprising a case circumferential wall that is configured to support the tray circumferential wall and that defines a first coupling slit configured to be coupled with the first coupling protrusion, wherein the second tray is configured to move between (i) a closed position where the first tray and the second tray are in contact each other and (ii) an open position where the first tray and the second tray are separated each other, and wherein, in the closed position, the first chamber and the second chamber are configured to define an ice chamber, and at least a part of the chamber wall of the first tray is configured to be positioned in a space defined by the tray circumferential wall of the second tray.
2. The ice maker of claim 1, wherein at least a part of the outer surface of the tray circumferential wall and at least a part of an inner surface of the case circumferential wall face each other.
3. The ice maker of claim 1, wherein the first coupling slit is recessed in the first direction, and the first coupling protrusion is inserted into the first coupling slit.
4. The ice maker of claim 3, wherein the first coupling protrusion comprises a neck part having a smaller diameter than a remaining part of the first coupling protrusion, the neck part being inserted into the first coupling slit.
5. The ice maker of claim 1, wherein the tray circumferential wall comprises a first linear portion

extended linearly from the tray body and a first curved portion curved away from the tray body, and the case circumferential wall comprises a second linear portion configured to surround the first linear portion and a second curved portion configured to surround the first curved portion, and wherein the first coupling protrusion is disposed at the first linear portion, and the first coupling slit is formed at the second linear portion.

6. The ice maker of claim 1, wherein the case further comprises a case plate in which an opening is formed, wherein at least a part of the tray circumferential wall passes through the opening, and wherein the case circumferential wall extends around a periphery of the opening.

7. The ice maker of claim 6, wherein the second tray further comprises an extension part extended outward from an interface between the tray body and the tray circumferential wall, and coupled to the case plate.

8. The ice maker of claim 7, wherein a first surface of the extension part contacts a first surface of the case plate, wherein the tray circumferential wall extends in the second direction from the first surface of the extension part, and wherein the case circumferential wall extends in the second direction from a second surface of the case plate opposite to the first surface of the case plate.

9. The ice maker of claim 8, further comprising: a support that is contact with a second surface of the extension part opposite to the first surface of the extension part, the support being configured to support the second tray.

10. The ice maker of claim 9, wherein the support defines a chamber accommodation portion that is configured to surround and support to the tray body.

11. The ice maker of claim 1, wherein the tray circumferential wall further comprises a second coupling protrusion protruded from the outer surface of the tray circumferential wall, and the case circumferential wall a second coupling slit configured to be coupled with the second coupling protrusion, wherein the first coupling protrusion and the second coupling protrusion are located on opposite sides of the tray circumferential wall, and wherein the first coupling slit and the second coupling slit are located on opposite sides of the case circumferential wall.

12. The ice maker of claim 11, wherein the tray circumferential wall comprises a first linear portion extended linearly from the tray body and a first curved portion curved away from the tray body, and the case circumferential wall comprises a second linear portion configured to surround the first linear portion and a second curved portion configured to surround the first curved portion, wherein the first coupling protrusion is disposed at the first linear portion, and the first coupling slit is formed at the second linear portion, and wherein the second coupling protrusion is disposed at the first curved portion, and the second coupling slit is formed at the second curved portion.

13. The ice maker of claim 1, wherein, in the closed position, a first contact surface of the first tray and a second contact surface of the second tray are contact with each other, wherein the first contact surface is located at an edge of the first chamber, and the second contact surface is located at an edge of the second chamber, and wherein the tray circumferential wall extends in the second direction from the second contact surface.

14. The ice maker of claim 1, wherein the first tray and the second tray are made of a flexible material.

15. A refrigerator comprising: a cabinet defining a storage space; and an ice maker provided in the storage space, wherein the ice maker comprises: a first tray that defines a first chamber, the first tray comprising a chamber wall defining the first chamber, a second tray that defines a second chamber, the second tray comprising (i) a tray body defining the second chamber recessed in a first direction, (ii) a tray circumferential wall extending from the tray body in a second direction opposite to the first direction, (iii) a first coupling protrusion that protrudes from an outer surface of the tray circumferential wall, and a case configured to support the second tray, the case comprising a case circumferential wall that is configured to support the tray circumferential wall and that defines a first coupling slit configured to be coupled with the first coupling protrusion, wherein the second tray is configured to move between (i) a closed position where the first tray and the second

tray are in contact each other and (ii) an open position where the first tray and the second tray are separated each other, and wherein, in the closed position, the first chamber and the second chamber are configured to define an ice chamber, and at least a part of the chamber wall of the first tray is configured to be positioned in a space defined by the tray circumferential wall of the second tray.

16. The refrigerator of claim 15, wherein at least a part of the outer surface of the tray circumferential wall and at least a part of an inner surface of the case circumferential wall face each other.

17. The refrigerator of claim 15, wherein the first coupling slit is recessed in the first direction, and wherein the first coupling protrusion comprises a neck part having a smaller diameter than a remaining part of the first coupling protrusion, the neck part being inserted into the first coupling slit.

18. The refrigerator of claim 15, wherein the case further comprises a case plate in which an opening is formed, wherein at least a part of the tray circumferential wall passes through the opening, and wherein the case circumferential wall extends around a periphery of the opening.

19. The refrigerator of claim 18, wherein the second tray further comprises an extension part extended outward from an interface between the tray body and the tray circumferential wall, wherein a first surface of the extension part contacts a first surface of the case plate, wherein the tray circumferential wall extends in the second direction from the first surface of the extension part, and wherein the case circumferential wall extends in the second direction from a second surface of the case plate opposite to the first surface of the case plate.

20. The refrigerator of claim 15, wherein the tray circumferential wall further comprises a second coupling protrusion protruded from the outer surface of the tray circumferential wall, and the case circumferential wall a second coupling slit configured to be coupled with the second coupling protrusion, wherein the first coupling protrusion and the second coupling protrusion are located on opposite sides of the tray circumferential wall, and wherein the first coupling slit and the second coupling slit are located on opposite sides of the case circumferential wall.
